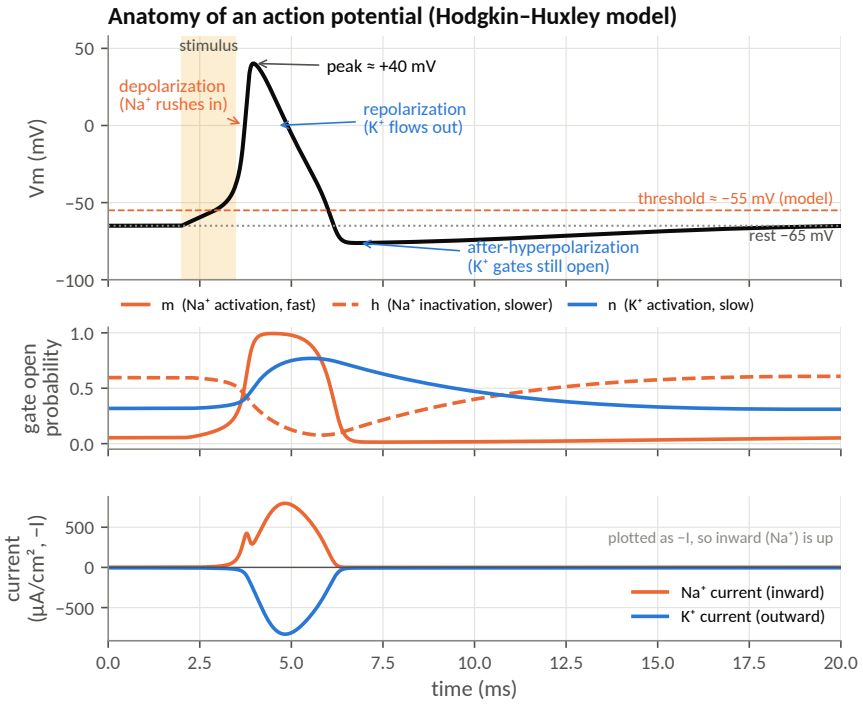
FIELD NOTES, EXPANDED

Electrophysiology from the bench up

Slices, patch clamp, synapses and the physics behind them

A study guide built on the handwritten notes from the SCA1 electrophysiology sessions (undated session · Sep 10, 2026 · Sep 21, 2026)



Prepared for Omkar · Department of Neurology, Northwestern University · September 2026
Contains 9 embedded animations (play in Adobe Acrobat Reader) and a companion set of MP4 videos.

How to use this guide

The **headings and their order follow your notebook** page by page. Each section has four layers:

1. **From your notes**, in the grey box: a thumbnail of the notebook page and a transcription of what you wrote. Words I could not read with confidence are marked [like this?].
2. **Explanation**: the concept in plain language, starting from first principles.
3. **Figures and animations**: simulated traces that show what the concept looks like on a real rig. Animations have ▶ controls underneath them.
4. **Coloured boxes**:
   - Key idea — the one thing to remember;
   - A gentle correction — where a note needs a small fix;
   - Check in the lab — a reading or rule worth confirming;
   - Going deeper — extra depth;
   - Connection to SCA1 — links to your own research.

Colour code used in every figure and animation: K^+ blue, Na^+ orange, Ca^{2+} green, Cl^- / GABA / NMDA violet, glutamate / AMPA magenta, Mg^{2+} amber.

If you only have 20 minutes, read Sections 4 (voltage vs current clamp), 8 (reversal potential), 28 (Nernst) and 34 (cesium internals), and watch videos 03, 01 and 08.

> **The three ideas everything else rests on** (1) **Ohm's law**: V = IR, so I = V/R. (2) Every channel's current is $g\,(V - E_{\text{rev}})$, and its sign depends on which side of E_{rev} you are. (3) **Inward** current (positive charge entering) is **negative** and plotted downward.

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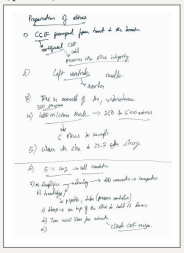
PART I

Slice preparation, the rig and the core physics

These pages follow the first block of your notes (pages 1–10). The order and headings match the notebook; each section starts with what you wrote, then adds the background you need to understand it.

1 Preparation of slices

From your notes (p. 1)



1) CSF pumped from heart to the brain → artificial CSF, cold → preserves the slice integrity.
2) Left ventricle, needle → aorta.
3) Fix in mount of the vibratome.
4) 400 [corrected to] 300 microns thick → 250 to 500 microns → 5 slices to sample.
5) Warm the slice to 33.7 after slicing.
6) 5% CO_2 → cell incubator.

The goal of slice preparation is to get living brain tissue from the animal into a dish *with the neurons still healthy and the local circuit still intact*. Everything in the protocol is about fighting three enemies: lack of oxygen (hypoxia), over-excitation from released glutamate (excitotoxicity), and mechanical damage from the blade.

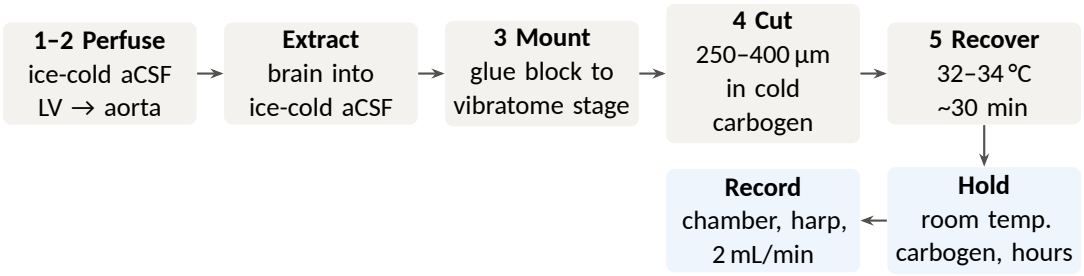


Figure 1: The acute-slice workflow your steps 1–6 describe.

1.1 Transcardial perfusion with cold aCSF (steps 1–2)

Under deep anaesthesia the chest is opened, a needle goes into the **left ventricle**, and the right atrium is snipped so fluid can leave. The heart's own anatomy then does the work: the left ventricle pumps into the **aorta**, and from there the solution reaches the whole body, including the brain. That is what "CSF pumped from heart to the brain" means. In reality it is *artificial* CSF (aCSF), pushed by a pump or gravity through the vasculature.

Why do this instead of just taking the brain out?

- **Cold** (0–4 °C) slows metabolism, so neurons consume less O_2 and ATP during the minutes without blood flow, and it reduces swelling.
- **Oxygenated** aCSF reaches deep tissue through the capillaries far faster than it could diffuse in from the surface.
- It **washes out blood** and replaces the extracellular fluid with a protective "cutting" solution. In many labs this is a modified aCSF with low Ca^{2+}/high Mg^{2+}, or with Na^+ replaced by sucrose or NMDG, which reduces excitotoxic damage while cutting (see Section 30).

1.2 Mounting and cutting on the vibratome (steps 3–4)

The brain (or the cerebellum block) is glued with cyanoacrylate onto the specimen stage, the "mount", often backed by an agar block. It is then submerged in ice-cold, carbogenated solution. A vibratome's blade oscillates side-to-side while advancing slowly, so it *saws* rather than presses. This is gentler than a microtome and works on unfixed tissue.

Thickness trade-off. You corrected 400 to 300 µm and noted a usable range of 250–500 µm. Thicker slices keep more dendrites and connections intact but oxygenate poorly in the core and are harder to see through. Thinner slices are healthier throughout, but more cells near the surface have cut dendrites. 250–300 µm is the common compromise for patch clamp. "5 slices to sample" is simply how many good sections you get from the region of interest.

> **Key idea** For **cerebellum**, slices are usually cut *sagittally* (parasagittal). Purkinje-cell dendrites are flat fans lying in the sagittal plane, so this orientation keeps the whole dendritic tree inside the slice (see Fig. 11).

1.3 Recovery at ~34 °C (step 5)

After cutting, slices go into carbogenated aCSF at 32–35 °C (your note: 33.7 °C) for about 30 minutes, then sit at room temperature until use. Warming lets ion pumps restore gradients, reseals damaged membranes, and lets slices "settle" before recording. Recording itself is usually done at room temperature or 30–34 °C. Always note which, because channel kinetics speed up roughly 2–3× per 10 °C (Q_{10}).

1.4 Why 5% CO_2? (step 6)

aCSF uses bicarbonate ($NaHCO_3$) as its pH buffer, the same system blood uses:

$$CO_2 + H_2O \rightleftharpoons H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$$

The CO_2 in the gas sets how much acid there is; the HCO_3^- (about 25 mM) sets the base. With 5% CO_2 the Henderson–Hasselbalch equation gives

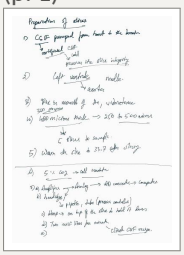
$$\mathrm{pH} = 6.1 + \log_{10} \frac{[HCO_3^-]}{0.03 \times P_{CO_2}} \approx 6.1 + \log_{10} \frac{25}{0.03 \times 36\,\mathrm{mmHg}} \approx 7.4.$$

So **carbogen does two jobs**: 95% O_2 keeps tissue alive, 5% CO_2 holds pH at 7.4. Stop bubbling and the solution

drifts alkaline within minutes. The same logic explains the **5% CO_2 cell-culture incubator**: culture media are also bicarbonate-buffered. That link is what your note "5% CO_2 → cell incubator" is pointing at.

2 The rig: from the cell to the computer

From your notes (p. 1)



7a) Amplifier → analog → AD converter → computer.
b) Headstage → pipette, tube (pressure controller).
c) Harp → on top of the slice to hold it down.
d) Two millilitres per minute. *e) Cited CSF recipe.*

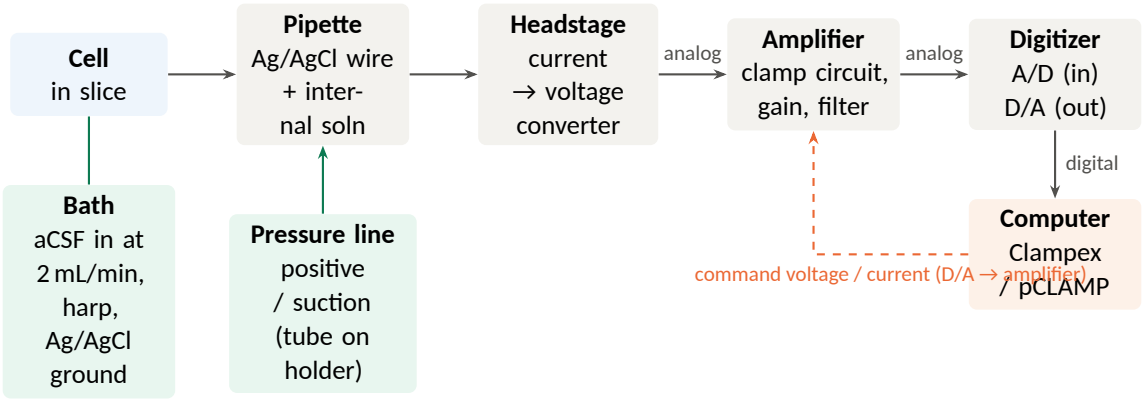


Figure 2: Signal chain. The tiny current flowing through the pipette (pA–nA) is converted to a voltage in the headstage, conditioned by the amplifier (e.g. low-pass Bessel filter at 2–10 kHz), digitized, and stored. The computer also sends the *command* (the voltage step or current step) back through the D/A converter.

Headstage
The small box mounted on the micromanipulator right next to the pipette holder. It contains the first stage of the amplifier (a high-value feedback resistor that turns current into voltage). It sits close to the cell because very small currents are easily swamped by noise over long cables.

Amplifier
Runs the voltage- or current-clamp feedback loop (Section 4), sets gain, and filters. The output is an *analog* voltage proportional to current (VC) or membrane voltage (CC).

A/D converter (digitizer)
Samples that analog signal at a fixed rate (you use 20 kHz; Section 14) and hands numbers to the computer. It also works in reverse (D/A) to deliver your protocol's command waveform.

Pressure controller / tube
Connected to the side port of the pipette holder. Positive pressure keeps the tip clean while approaching; gentle suction forms the seal; a brief strong pulse breaks in (see Animation 4).

Harp

A U-shaped platinum (or stainless) frame with nylon threads stretched across it (your drawing, p. 26). It lies on the slice so the flowing aCSF doesn't move it. Position the threads away from the cells you want to patch.

Perfusion, 2 mL/min

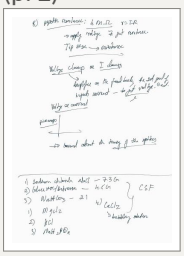
Fresh, carbogenated aCSF flows continuously through the chamber (typically 1–4 mL/min, often via an in-line heater). It supplies O_2, removes waste, and lets you *wash in* drugs. Keep the flow steady: fluctuations change the bath level, which changes pipette capacitance and adds noise.

Ag/AgCl electrodes

The chlorided silver wire in the pipette and the bath ground ("silver chloride", p. 18) convert ionic current into electron current: $AgCl + e^- \rightleftharpoons Ag + Cl^-$. Both solutions therefore need Cl^-. Re-chloride the wire when the grey coating wears off, or offsets drift.

3 Pipette resistance

From your notes (p. 2)



8) Pipette resistance: [6?] MΩ. V = IR → apply voltage to get resistance. Tip size → resistance.

The glass pipette filled with salt solution is a resistor. Its narrowest part, the tip, dominates the resistance. To measure it, the amplifier applies a small, known voltage step (the **seal test**, usually −5−−10 mV) and measures the current. Ohm's law then gives

$$R = \frac{\Delta V}{\Delta I} = \frac{5\ \text{mV}}{0.83\ \text{nA}} \approx 6\ \text{MΩ}.$$

Your seal-test screenshot (p. 16) shows exactly this: pulse −5 mV, holding 0 mV, and a readout of 8.7 MΩ. That predicts a current step of $-5/8.7 \approx -0.57$ nA (−575 pA), which matches the square in the photo.

Key idea **Units trick:** mV / MΩ = nA, and mV / GΩ = pA. Memorize this; it makes every V = IR calculation at the rig instant.

Tip size ↔ resistance. A smaller opening means higher resistance. Typical whole-cell pipettes are 3–7 MΩ. Lower resistance (bigger tip) gives lower access resistance and better voltage control, but seals are harder. Higher resistance (smaller tip) seals more easily but gives higher access resistance and slower, noisier recordings. Purkinje cells, with their huge capacitance, are usually patched with lower-resistance pipettes. Cultured neurons tolerate 4–7 MΩ.

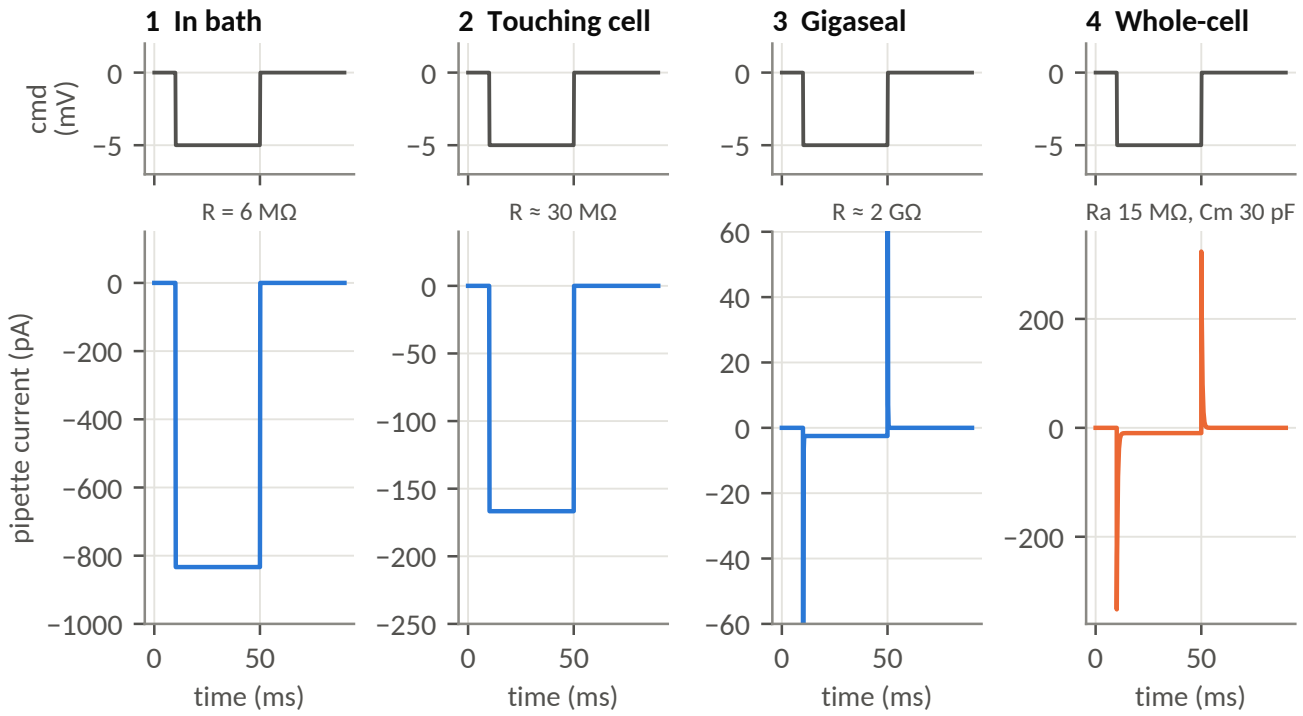


Figure 3: Simulated seal-test response through the four stages of getting a recording. Same −5 mV pulse in every panel. What changes is the resistance the current has to flow through (note the different y-axes).

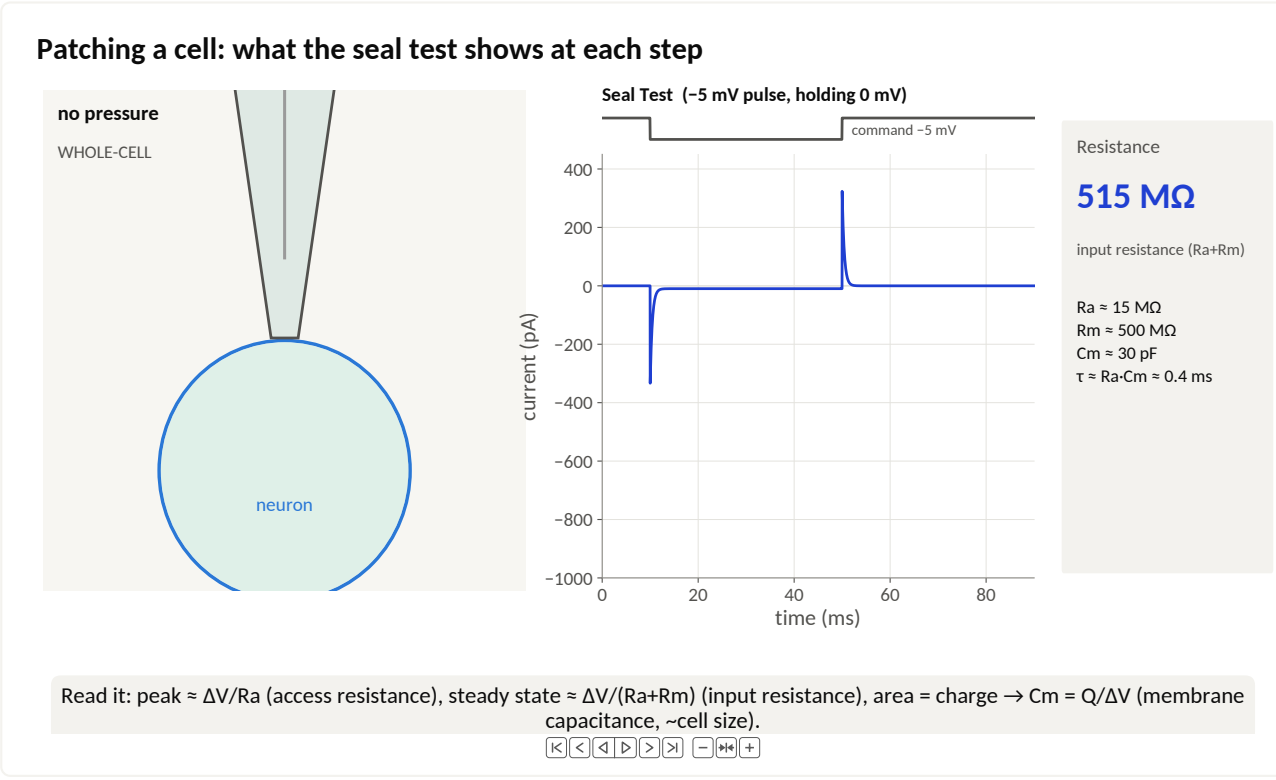
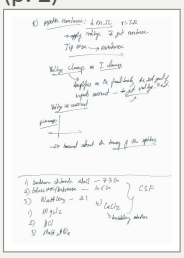


Figure 4: Animation — From bath to whole-cell. The pipette approaches with positive pressure, touches, seals (resistance climbs past 1 GΩ), then breaks in. Watch the readout and the current trace on the right. *Press ▶ below the frame (Adobe Acrobat Reader / Okular); other viewers show the final frame. Same animation as video `04_seal_test_breakin.mp4`.*

4 Voltage clamp or current clamp

From your notes (p. 2)



Voltage clamp or I clamp. Amplifier in the feedback: the set goal of voltage, injects current to get voltage. Voltage or current. [axis labelled] picoamps. → Learned about the timing of the spikes.

These are the two basic modes of whole-cell recording. They are opposites: in each, you *control* one variable and *measure* the other.

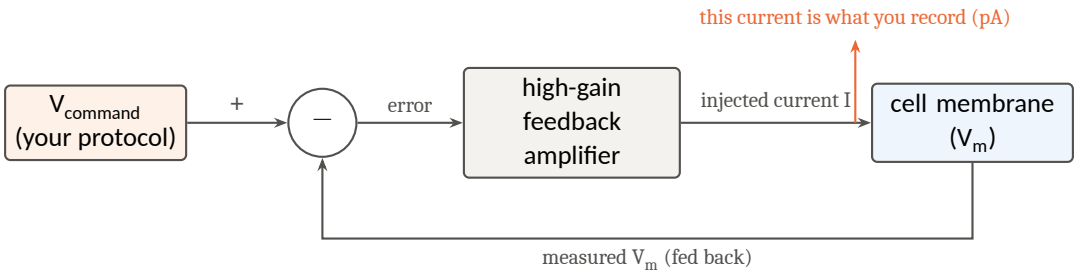


Figure 5: Voltage clamp as a negative-feedback loop. The amplifier continuously compares the measured V_m with the command and injects whatever current removes the difference. When channels open and let current flow, the amplifier must supply an equal and opposite current to keep V_m still, so *the injected current is a mirror of the membrane current*.

	Voltage clamp (VC)	**Current clamp (CC)**
You set	membrane voltage (e.g. hold at −70 mV, steps)	injected current (0 pA, or steps)
You measure	current, in **pA** (your "picoamps" axis)	voltage, in **mV**
Answers	"How are the *receptors / channels* doing?" (p. 16)	"How is the *cell* doing?" (p. 16)
Typical use	EPSCs/IPSCs, I–V curves, channel kinetics, pharmacology	resting potential, APs, firing patterns, F–I curves, EPSPs
Signals look like	inward current = downward deflection	depolarization = upward deflection

"Learned about the timing of spikes" fits current clamp: when the voltage is free to move, you see *when* the cell fires relative to its inputs. In voltage clamp, spikes are (ideally) prevented.

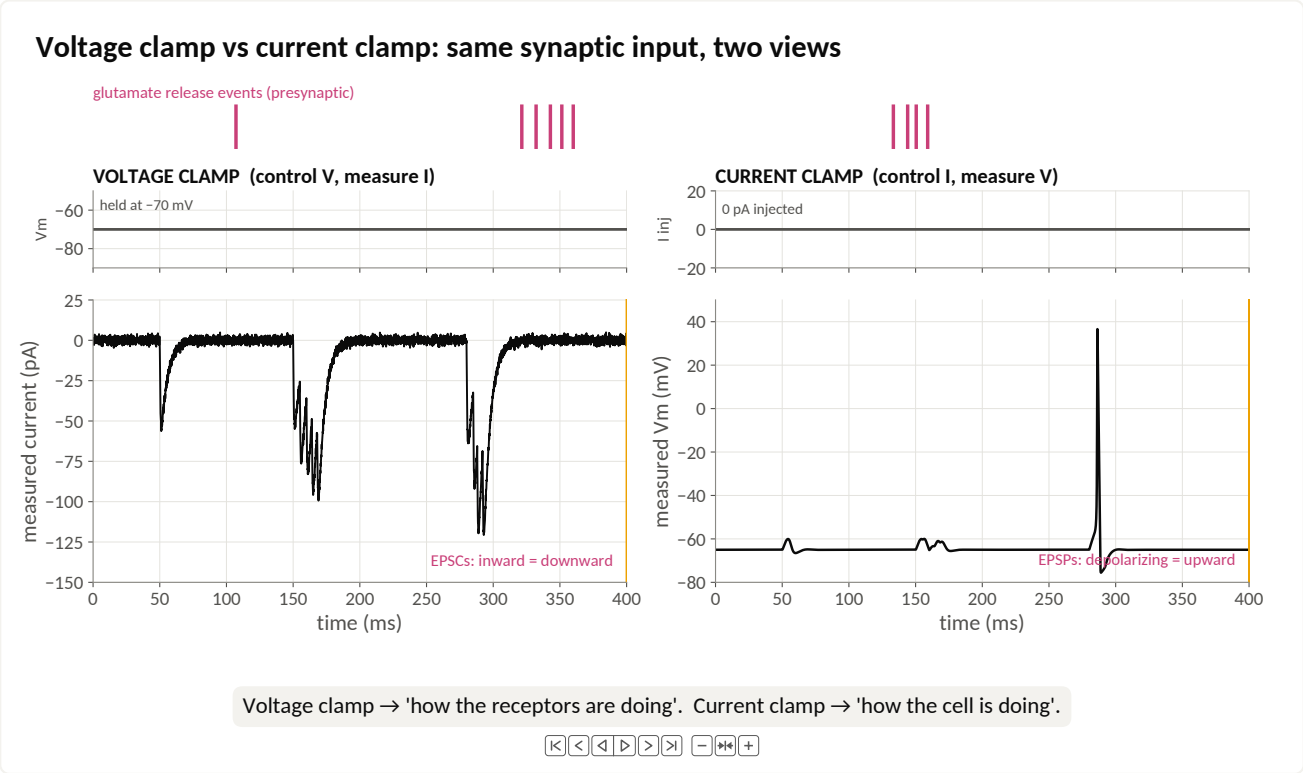
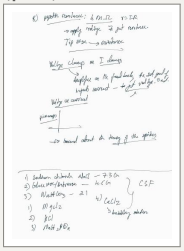


Figure 6: Animation — Voltage clamp vs current clamp. The same glutamate release events (top) seen both ways. Left: the voltage is flat and the current shows EPSCs. Right: the injected current is flat and the voltage shows EPSPs, and a burst drives a spike. *Press ▶ below the frame (Adobe Acrobat Reader / Okular); other viewers show the final frame. Same animation as video `03_voltage_vs_current_clamp.mp4`.*

5 The aCSF recipe

From your notes (p. 2)



1) Sodium chloride NaCl — 7.3 g. 2) Glucose/dextrose — 4.[6?] g. 3) $NaHCO_3$ — 2.1. 4) $CaCl_2$ → bubbling solution. Plus: 1) $MgCl_2$, 2) KCl, 3) NaH_2PO_4. [all → CSF]

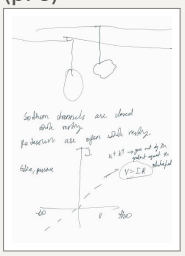
Your three weighed amounts are exactly a standard **1-litre** aCSF. Converting grams to millimolar (mM = g/L ÷ molecular weight × 1000):

Component	MW (g/mol)	Your note (g/L)	Concentration	Role
NaCl	58.44	7.3	**125 mM**	main osmolyte; sets Na^+ and Cl^- gradients
Glucose (anhydrous)	180.16	4.5–4.6	**25 mM**	energy (high, for slice metabolism)
$NaHCO_3$	84.01	2.1	**25 mM**	pH buffer with 5% CO_2
Typical values for the rest (check your lab's cited recipe):				
KCl	74.55	0.19	2.5 mM	sets K^+ gradient → resting potential
NaH_2PO_4	119.98	0.15	1.25 mM	phosphate
$MgCl_2{\cdot}6H_2O$	203.30	0.20	1 mM	stabilizes membranes; NMDA Mg^{2+} block
$CaCl_2{\cdot}2H_2O$	147.01	0.29	2 mM	synaptic release, channel gating

Why "$CaCl_2$ → bubbling solution" Add $CaCl_2$ **last**, into solution that is already bubbling with carbogen. Without CO_2, the bicarbonate solution turns alkaline, and Ca^{2+} precipitates with carbonate and phosphate. The solution goes faintly cloudy, and you have silently lost calcium. The osmolarity of the finished aCSF should be about 295–310 mOsm; check it with an osmometer.

6 Which channels are open at rest? Glia vs neurons

From your notes (p. 3)



[Drawing: two pipettes on two cells.] Sodium channels are closed while resting. Potassium are open while resting. Glia, passive. [I–V: straight line through –80] V = IR. K^+ K^+ → goes out by the gradient against the electric field.

At rest, a neuron's membrane is mostly permeable to K^+ (leak and inward-rectifier channels are open) and only slightly to Na^+ (Na^+ channels are closed). K^+ is concentrated inside, so it leaks *out* down its concentration gradient. Each ion that leaves takes positive charge with it, making the inside negative. That negativity then pulls K^+ back. The two forces balance near the K^+ equilibrium potential. This is the resting potential, explained fully with numbers in Section 28.

Glia are "passive". Astrocytes are dominated by K^+ conductances (Kir4.1, K2P channels) and have almost no voltage-gated channels. Their I–V relation is therefore a straight line: Ohm's law, V = IR, with the slope equal to the conductance $g = 1/R$. The line crosses zero current near E_K (−80–−90 mV). That is why an astrocyte sits so negative, and why your drawing crosses near -80. Low input resistance (a few tens of MΩ) and no spikes are the quickest way to recognise that you have patched a glial cell by mistake.

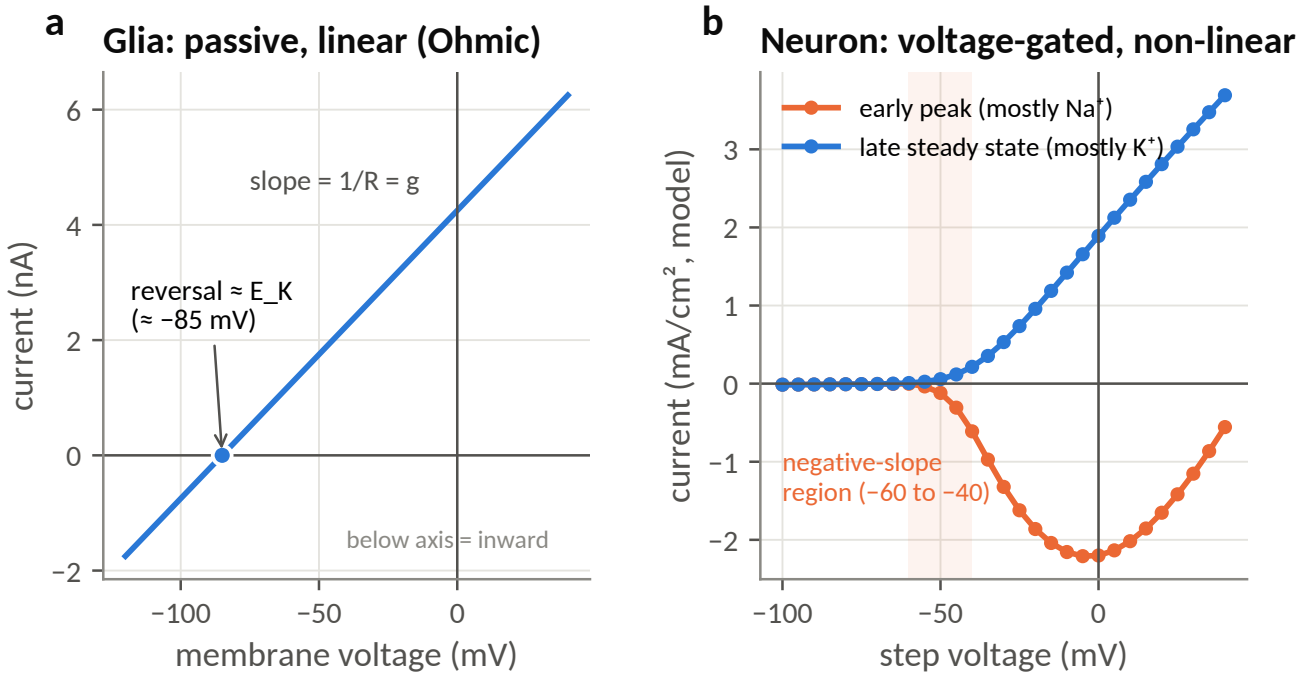
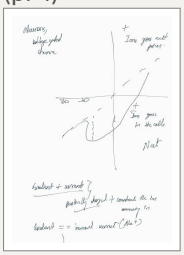


Figure 7: **a**, A passive (glial) membrane: current is proportional to voltage (Ohmic) and reverses near E_K. **b**, A neuron (Hodgkin–Huxley model): voltage-gated channels make the I–V curve bend. The early transient current (orange) becomes *inward* and grows as the cell is depolarized past about −55 mV, giving the **negative-slope region** you drew on p. 4. The late current (blue) is outward K^+.

7 Neurons: voltage-gated channels and the non-linear I–V

From your notes (p. 4)



Neurons, voltage-gated channels. [I–V with a dip between −60 and −40.] + ions go out: positive. Ions go in the cells (negative): Na^+. Gradient + current → positively charged, + counteracts the ions coming in. Gradient == inward current (Na^+).

Your drawing is the classic neuron I–V curve (Fig. 7b):

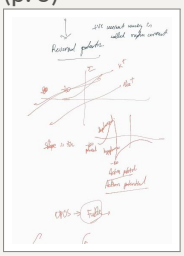
- Around rest (−80–−60 mV) it looks like glia: small, roughly linear leak currents.
- From about −55 mV, voltage-gated Na^+ channels start to open. Na^+ rushes *in* (down its concentration gradient *and* pulled by the negative interior), so the current becomes inward (negative). More depolarization opens more channels, so the curve goes *down* as voltage goes *up*. That is the negative slope, and it is the signature of the positive feedback that makes an action potential all-or-none.
- At more positive voltages the Na^+ current shrinks again. The inside is now positive and *counteracts* the ions

coming in ("+ counteracts the ions coming in"). It would reach zero at E_{Na}, where the electrical push exactly cancels the concentration gradient.

Sign convention. By convention, **positive charge moving into the cell is an inward current and is negative**, plotted downward. Positive charge leaving (or negative charge entering) is outward and positive. This single rule explains most of the signs in your notes.

8 Gradient, current and reversal potential

From your notes (p. 5)



→ Reversal potential. +ve current coming in called [inward?] current. [I–V lines for K^+ and Na^+, crossing zero at different voltages.] Slope is 1/R. –60 potential; depolarized / hyperpolarized; –80. Action potential.

Every ion channel type has a **reversal potential**, E_{rev}: the membrane voltage at which its net current is zero. Above E_{rev} the current flows one way, below it flows the other. For a channel that passes a single ion species, E_{rev} is that ion's Nernst (equilibrium) potential. For a channel that passes several ions (AMPA/NMDA pass Na^+ and K^+), it is a weighted average.

The current through a population of open channels is Ohm's law with a shifted zero:

$$I_{\mathrm{ion}} = g_{\mathrm{ion}}\,(V_m - E_{\mathrm{rev}}) \qquad \text{(slope } = g = 1/R\text{; the term in brackets is the } \textit{driving force}\text{)}$$

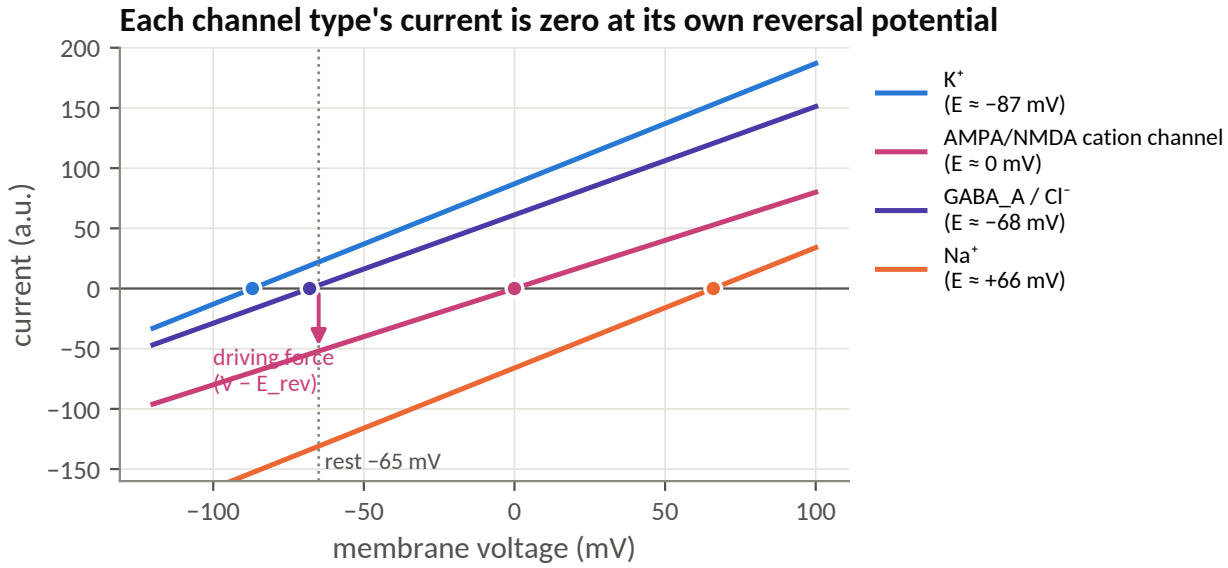


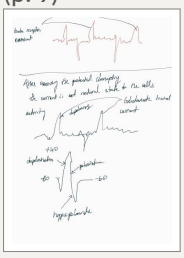
Figure 8: The I–V lines you drew on p. 5, drawn to scale for typical mammalian ion gradients at 37 °C. At rest (−65 mV, dotted), K^+ current is outward (above zero), while Na^+ and glutamate-receptor currents are inward (below zero). The arrow shows the driving force for the glutamate current at rest.

Depolarized vs hyperpolarized. Relative to rest (about −60 mV in your drawing): *depolarized* means less negative (toward 0 and beyond); *hyperpolarized* means more negative (toward −80 mV, E_K). Opening a channel pulls V_m toward that channel's E_{rev}. Opening Na^+ or glutamate channels therefore depolarizes; opening K^+ channels hyperpolarizes. Opening Cl^- channels clamps V_m near −70 mV.

The one rule that explains synapses A channel's effect depends only on where V_m is relative to that channel's E_{rev}. This is why GABA can be inhibitory in one cell and depolarizing in another (Section 33), and why holding the cell at a receptor's E_{rev} makes that receptor "disappear" from your recording (Section 34).

9 The action potential

From your notes (p. 9)



[AP drawing:] −60 → +40, depolarization, repolarization, hyperpolarization, back to −60. (See also p. 29: "Na^+ enters, K^+ go out of the cell, Na^+ channels will close", with separate Na^+ and K^+ current shapes.)

An action potential is a brief, stereotyped reversal of membrane voltage produced by two voltage-gated channel types acting on different timescales. Hodgkin and Huxley (1952) described it with three "gates":

- m: Na^+ **activation**. Opens fast when depolarized.
- h: Na^+ **inactivation**. Closes (plugs the channel) more slowly when depolarized. It must recover at negative voltages before the channel can open again, which creates the *refractory period*.
- n: K^+ **activation**. Opens slowly when depolarized.

Na^+ conductance $\propto m^3h$; K^+ conductance $\propto n^4$.

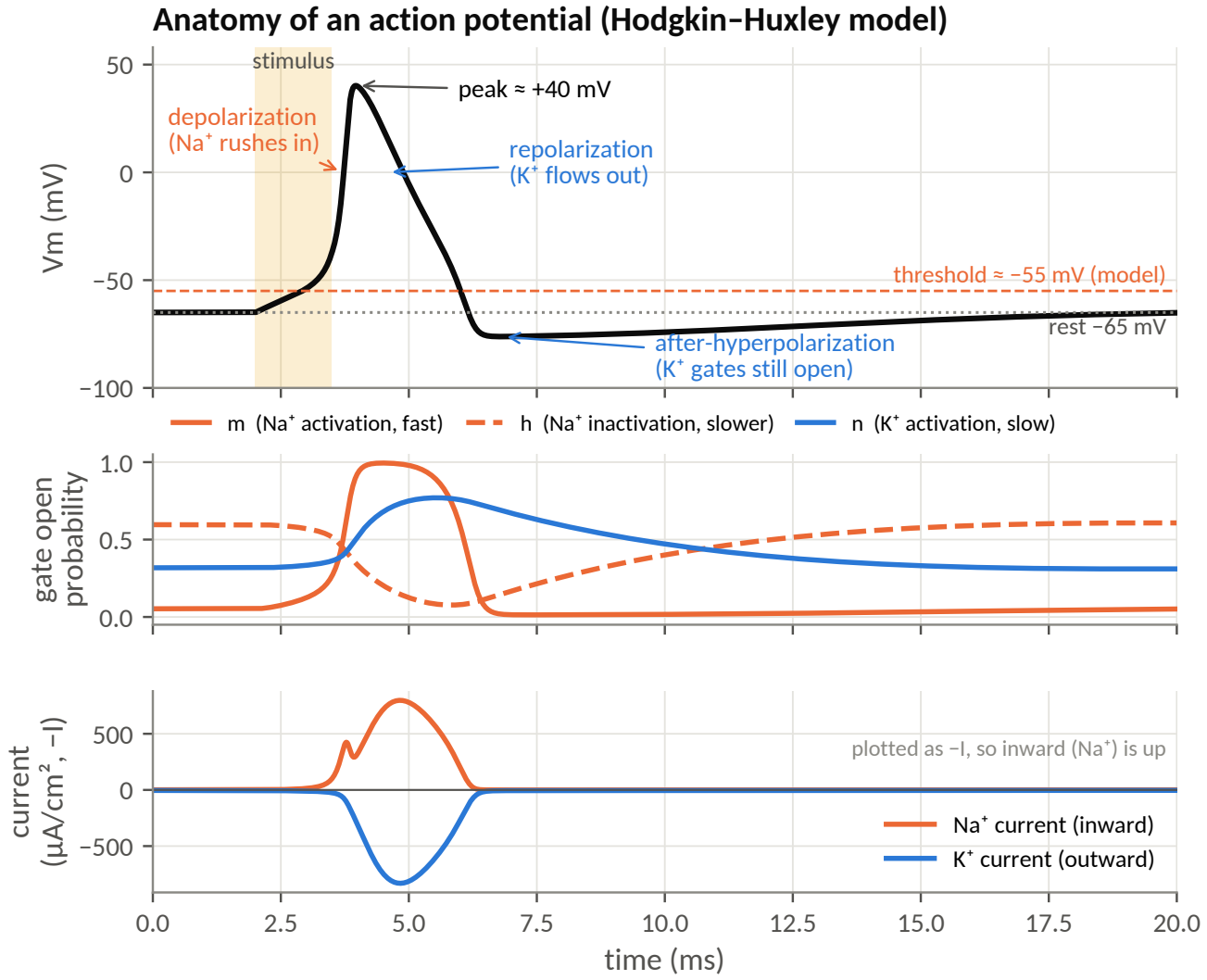


Figure 9: The five phases you labelled, simulated. The spike peaks near 40 mV (not at E_{Na}), because Na^+ channels inactivate and K^+ channels open while it is still rising. The after-hyperpolarization dips toward E_K because K^+ gates close slowly. Bottom panel: inward Na^+ current (plotted upward here) is brief and early; outward K^+ current is later and longer, matching your p. 29 sketch.

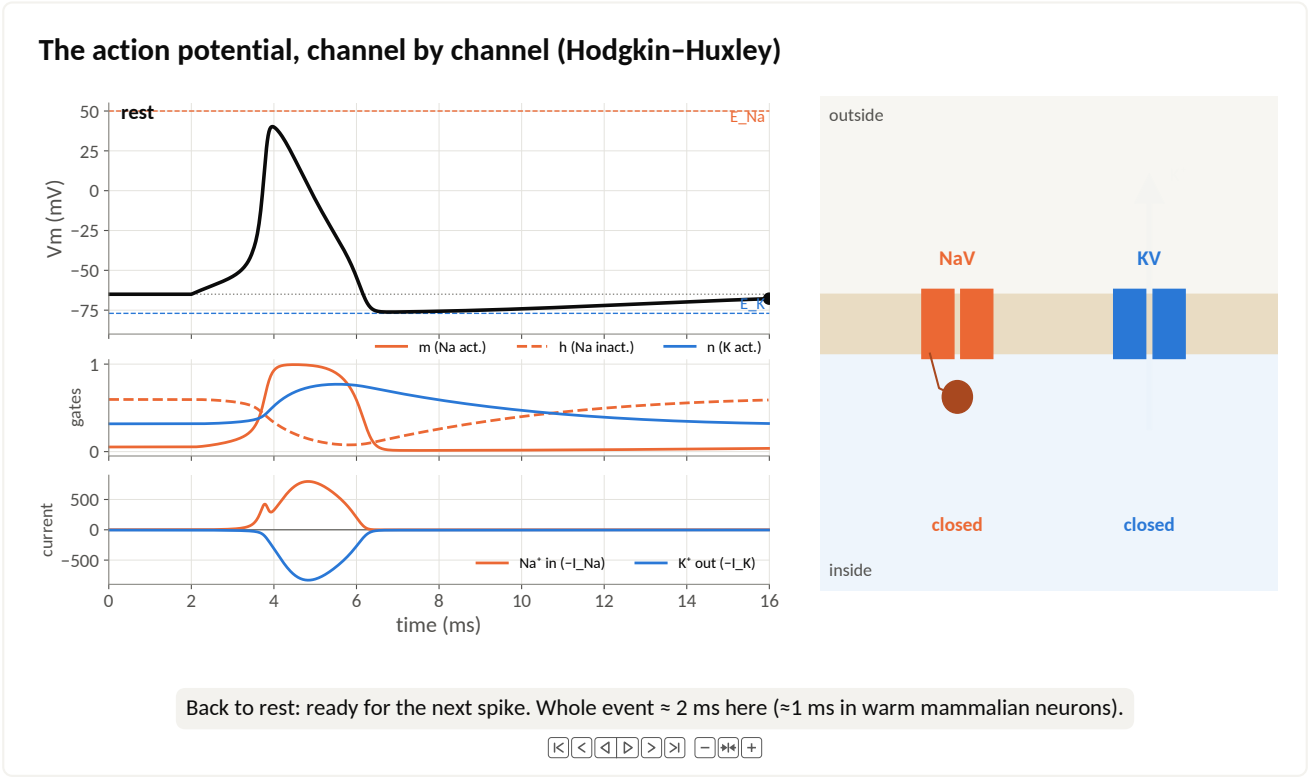
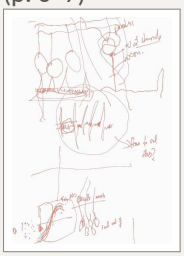


Figure 10: Animation — The action potential, channel by channel. Left: voltage, gates and currents. Right: a cartoon Na^+ channel (activation gate plus inactivation “ball”) and a K^+ channel. Watch the ball plug the Na^+ channel at the peak. *Press ▶ below the frame (Adobe Acrobat Reader / Okular); other viewers show the final frame. Same animation as video* `02_action_potential.mp4`.

10 The cerebellar circuit: “how to read this?”

From your notes (p. 6–7)



[Drawing of Purkinje cells, a layer, Na^+ channels at the axon, a recording “How to read this?”; “very old circuits works”, “read out of”.] Purkinje cells → checks out of [it?]. 10,000 granule → Purkinje → cerebellum. Climbing fibres connect to cortex.

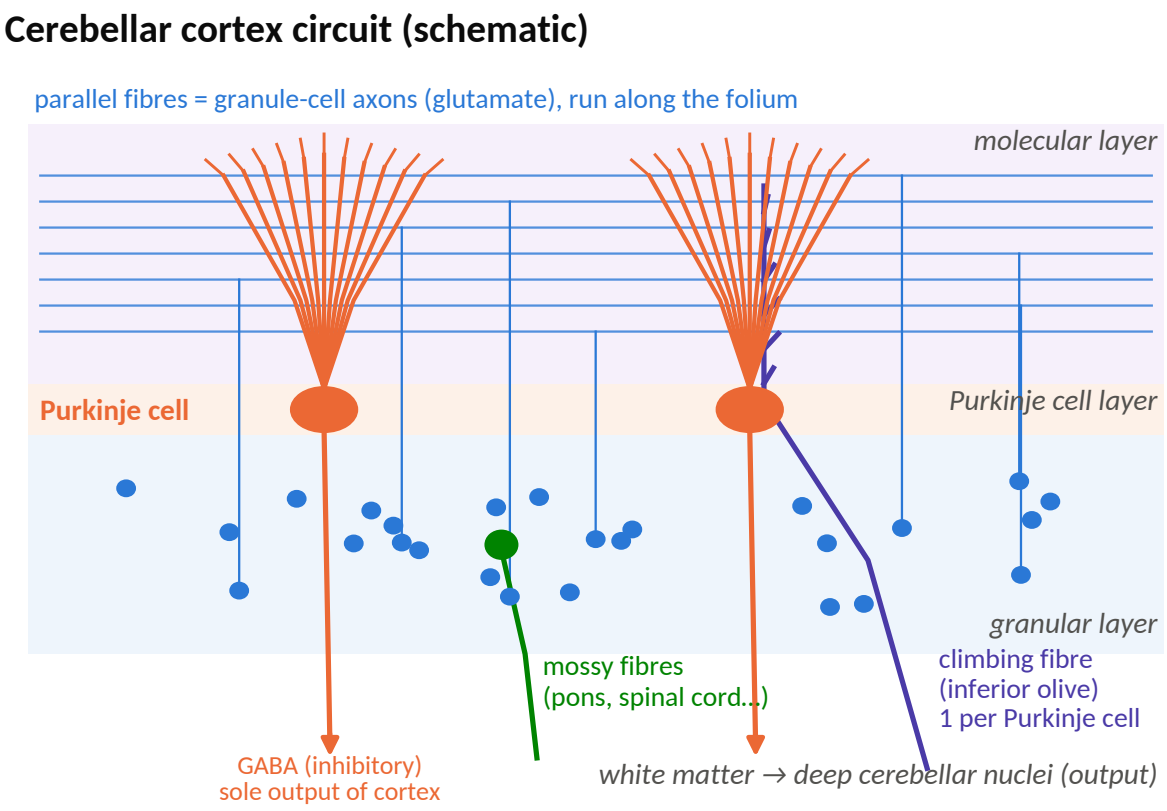


Figure 11: The cerebellar cortex has three layers and a strikingly regular, evolutionarily ancient wiring diagram (your “very old circuits”). Granule cells (blue) receive mossy fibres and send **parallel fibres** through the Purkinje dendrites. Each Purkinje cell also receives exactly one **climbing fibre** from the inferior olive. Purkinje axons are the *only output* of the cerebellar cortex and are inhibitory (GABA) onto the deep cerebellar nuclei.

How to read this circuit:

- **Purkinje cells are the read-out.** Everything the cerebellar cortex computes leaves through Purkinje axons (your “Purkinje cells → checks out”). They fire spontaneously at high rates (tens of Hz) even without input (intrinsic pacemaking). Information is carried by *pauses and changes* in that firing.
- **Two excitatory inputs, very different.** A Purkinje cell receives on the order of *100 000–200 000* parallel-fibre synapses in rodents, each weak. That is even more than the 10 000 in your note. They drive ordinary **simple spikes**. It receives one climbing fibre, which wraps the proximal dendrite with hundreds of contacts. One climbing-fibre input produces a large, all-or-none **complex spike**: a burst riding on a dendritic Ca^{2+} spike. When climbing-fibre and parallel-fibre inputs coincide, the parallel-fibre synapses weaken (PF-LTD, which needs mGluR1; Section 21). This is thought to underlie motor learning.
- **Where the Na^+ channels are.** Your drawing marks Na^+ channels at the axon initial segment. That is where spikes start. Purkinje dendrites have few Na^+ channels but many Ca^{2+} and K^+ channels.

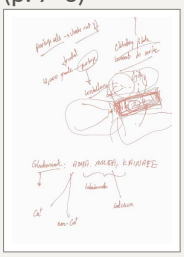
Connection to SCA1

Purkinje cells are among the earliest and most severely affected neurons in SCA1, so this circuit is where your electrophysiology and your snRNA-seq work meet.

- In ATXN1[82Q] mice, Purkinje pacemaker firing is normal at first. Around motor-symptom onset it becomes disrupted, with abnormal depolarization linked to reduced potassium-channel expression (BK channels and subthreshold K^+ channels). Later, dendritic atrophy partly *restores* firing, which the authors interpret as a compensatory shrinkage (Dell'Orco et al., 2015, *J Neurosci*).
- K^+-channel-targeting drugs have been tested in SCA1 mice. See *"Aminopyridines correct early dysfunction and delay neurodegeneration in a mouse model of spinocerebellar ataxia type 1"* (Hourez et al., 2011, *J Neurosci*), and *"Dendritic potassium channel dysfunction may contribute to dendrite degeneration in spinocerebellar ataxia type 1"* (PLoS One, 2018).
- Transcriptomic studies find a Purkinje-enriched gene module that declines with disease (Ingram et al., 2016, *Neuron*). Many ion-channel and Ca^{2+}-signalling genes that set the firing properties described in this guide sit in such modules. Ephys readouts that map directly onto them: spontaneous firing rate (cell-attached, Section 32), AP width and AHP (BK/SK), input resistance, and dendritic Ca^{2+} spikes.

11 Glutamate receptors: AMPA, NMDA, kainate

From your notes (p. 7–8)



Glutamate: AMPA, NMDA, KAINATE → glutamate. Ca^+ / non-Ca^+ / calcium. Metabotropic glutamate receptor: [SREPB?] [likely GPCR]. [Sketch: a curled structure — possibly hippocampus.]

Glutamate is the main excitatory transmitter in the brain. It acts on two families of receptors:

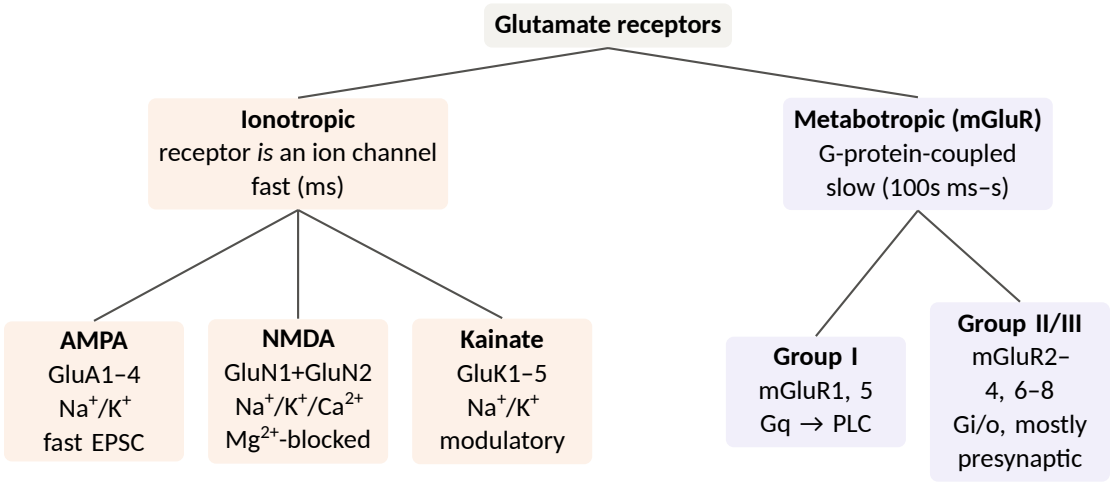


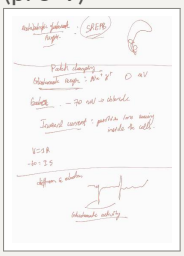
Figure 12: The glutamate receptor family tree.

The "Ca^{2+} / non-Ca^{2+}" split in your note. NMDA receptors always let Ca^{2+} through. AMPA receptors usually do *not*, because they contain the GluA2 subunit. Its RNA is edited at the "Q/R site", which puts a positively charged arginine in the pore that repels Ca^{2+}. AMPA receptors *lacking* GluA2 are Ca^{2+}-permeable and inwardly rectifying (blocked by intracellular polyamines at positive voltages). They are more common in immature neurons, which is what the paper on your screen (p. 22, developmental AMPAR changes and GluR2) is about.

The metabotropic receptors are covered in Section 21.

12 Patch clamping: which way do synaptic currents go?

From your notes (p. 8–9)



Patch clamping. Glutamate receptor: Na^+ K^+, 0 mV. GABA: –70 mV → chloride. Inward current: positive ions coming inside the cells. V = IR; –60 = I × 5. Diffusion & electrons. [Trace with a downward event: "glutamate activity".] [p. 9: GABA receptor current, upward events.]

This page brings everything together:

- **Glutamate receptors (AMPA/NMDA) reverse at about 0 mV**, because they pass Na^+ in and K^+ out about equally. At any negative holding potential, the net glutamate current is **inward**, so EPSCs are **downward** deflections (your "glutamate activity").
- **$GABA_A$ receptors pass Cl^-** and reverse near $E_{Cl} \approx -70$ mV (with normal, low intracellular Cl^-). Held positive to –70 mV, Cl^- flows *in*. Negative charge in counts as **outward** current, so IPSCs are **upward** (your p. 9 sketch).
- **Your V = IR example.** "$-60 = I \times 5$" gives $I = -12$ with units of mV / MΩ = nA, i.e. –12 nA if R is 5 MΩ.

That is huge. With a realistic neuronal input resistance of 500 MΩ, the same 60 mV corresponds to 120 pA. Always carry the units.

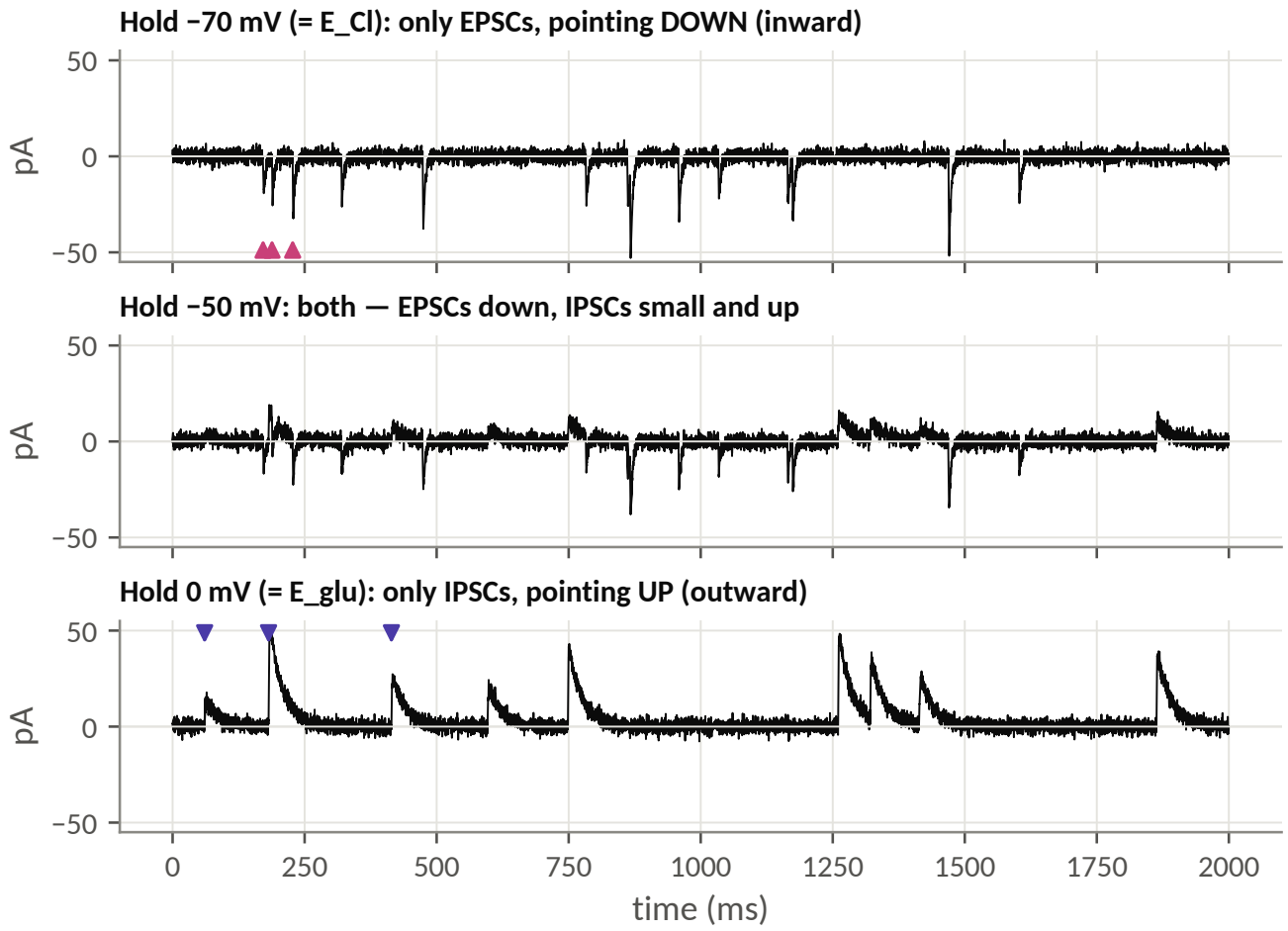
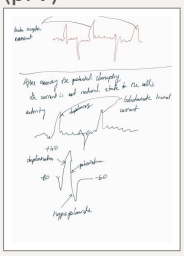


Figure 13: Simulated spontaneous synaptic activity at three holding potentials (low-Cl^- internal). Holding at a receptor's reversal potential removes that receptor's events from the trace.

13 Releasing the clamp: the same activity in current clamp

From your notes (p. 9)



After removing the potential clamping, the current is at natural state to the cells. Glutamate inward current activity → depolarizing.

Switching the amplifier to current clamp (I = 0) lets the membrane voltage move. The same glutamate receptor openings that were inward currents now *depolarize* the cell (EPSPs). Several EPSPs arriving close together *summate* in time. If they cross threshold, the cell fires.

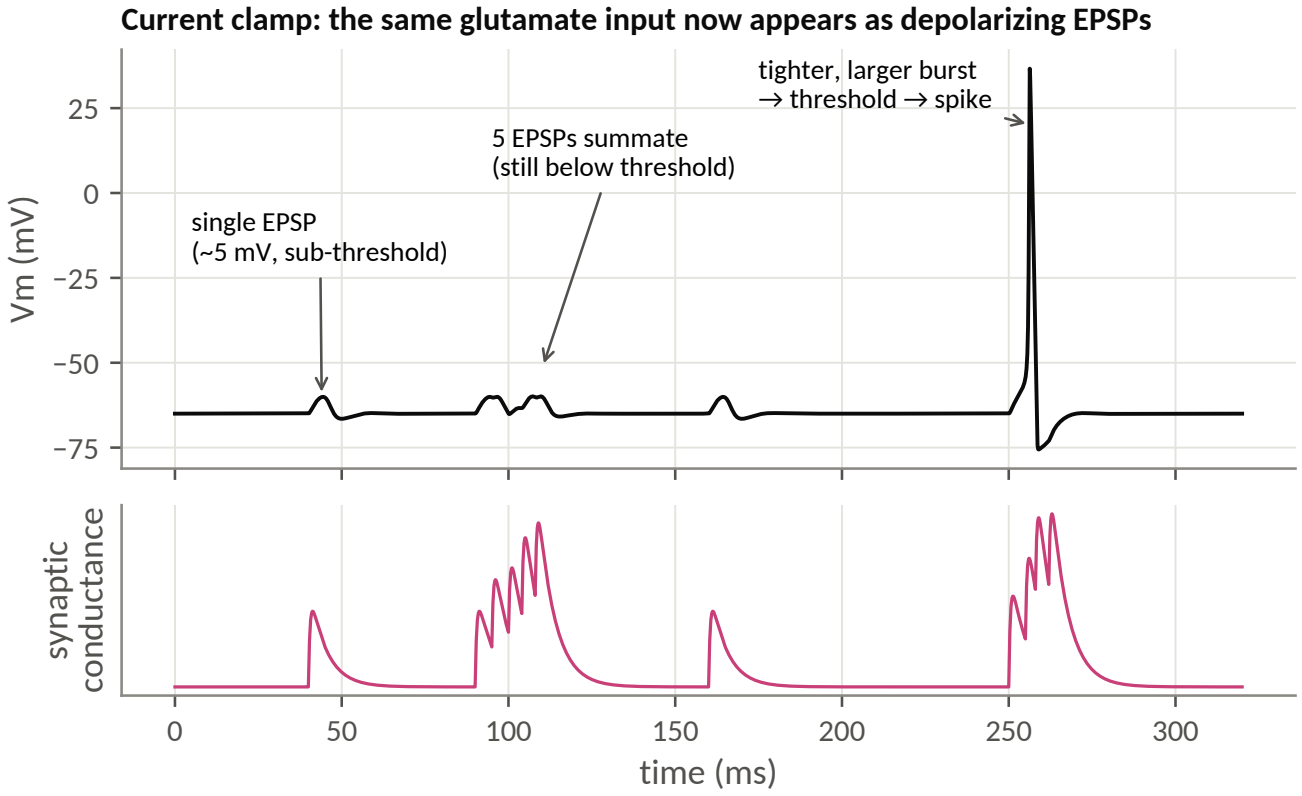
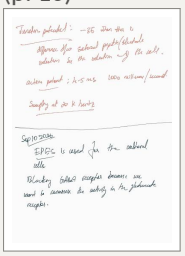


Figure 14: Current-clamp view of glutamatergic input (model). A single EPSP is a few mV. Five spread-out EPSPs summate but stay sub-threshold. A tighter, stronger burst reaches threshold and triggers a spike.

14 Junction potential, spike duration and sampling rate

From your notes (p. 10)



Junction potential: [−35?] when there is difference b/w external pipette (electrode) solution & the solution in the cell. Action potential: 1–5 ms; 1000 milliseconds / second. Sampling at 20 kHz.

14.1 Liquid junction potential (LJP)

Where two different solutions meet, here the pipette solution and the bath at the open tip, ions diffuse at different speeds. Gluconate is big and slow; Cl^- is small and fast. The resulting charge separation is a small voltage: the **liquid junction potential**. You zero the amplifier offset with the pipette in the bath, *while the LJP is present*. After you go whole-cell the junction effectively disappears, so every voltage you read is off by the LJP:

$$V_{\text{cell}} = V_{\text{measured}} - \text{LJP}$$

For typical K-gluconate internals against aCSF the LJP is about 15 mV. A measured resting potential of −50 mV is really about −65 mV. Many papers report whether they corrected for it. Calculators such as LJPcalc (Harden) compute it from your recipes.

> **Transcription check**
>
> Your note reads "−35". That is unusually large for a junction potential; typical LJPs are 5–16 mV. Worth confirming whether −35 mV referred to something else discussed at the same time, e.g. an **AP threshold**. Threshold is typically −55–−35 mV, and −35 mV is plausible in immature cultured neurons.

14.2 How long is a spike, and how fast must you sample?

Mammalian APs last about 0.3–2 ms at physiological temperature: Purkinje and fast-spiking interneurons are at the short end, pyramidal cells around 1 ms, and immature or cultured neurons can be broader. Your "1–5 ms" covers slow cells and room temperature.

Sampling at 20 kHz means one point every 50 µs, so 20+ points per millisecond-long spike. Two rules:

- **Nyquist**: you must sample at more than twice the highest frequency present, or fast events alias. In practice, sample 2–5× the low-pass filter cutoff (e.g. filter at 2–10 kHz, sample at 10–50 kHz).
- **Peak fidelity**: amplitude and half-width measurements need several points on the rising phase (Fig. 15).

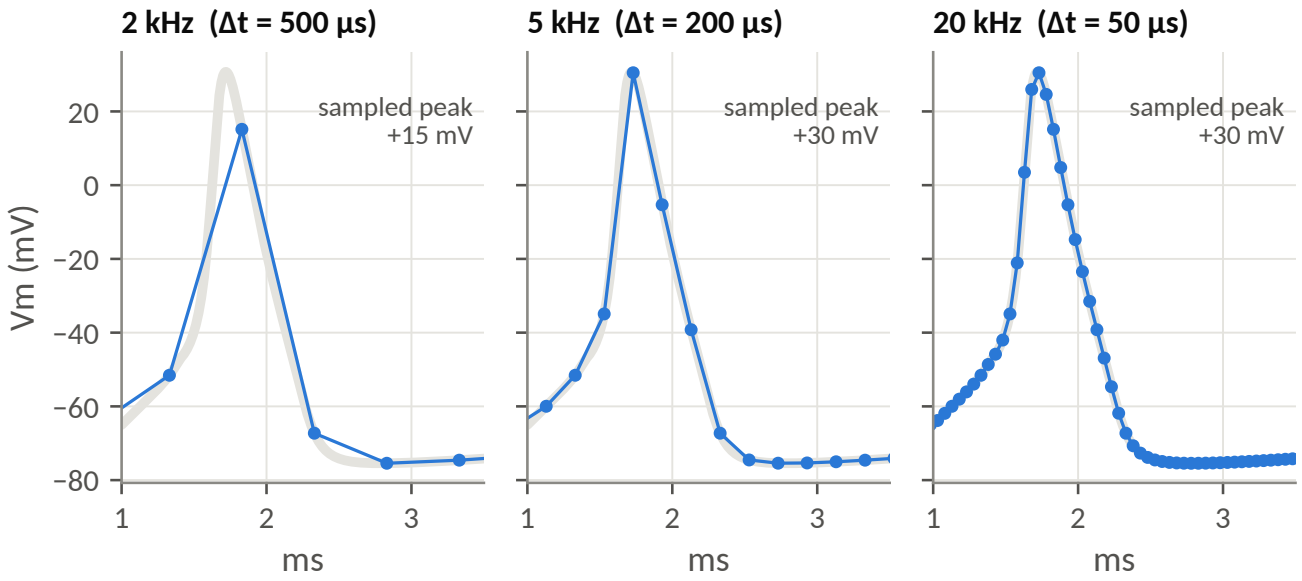


Figure 15: The same fast spike digitized at three rates. Grey: true signal; blue: recorded points. Low rates clip the peak and distort the shape.

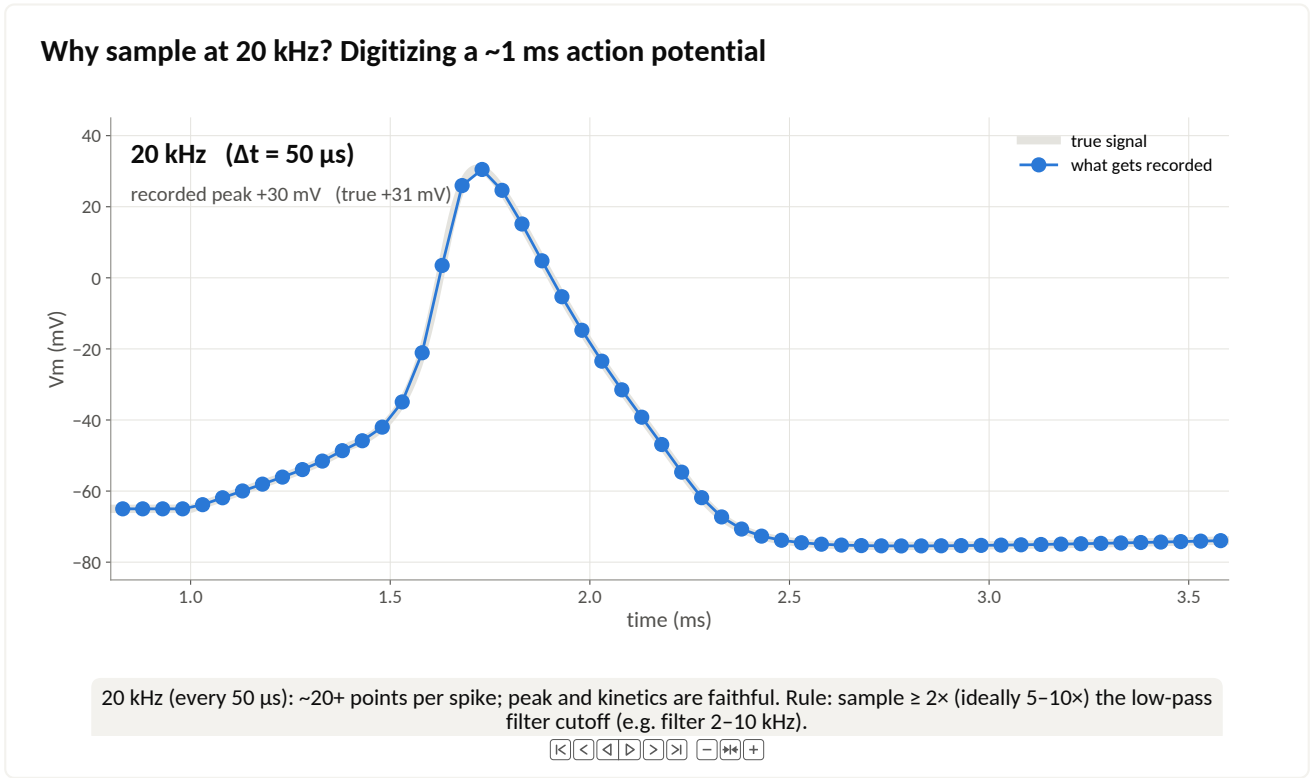


Figure 16: Animation — Sampling rate. Samples dropping onto a spike at 1, 2, 5 and 20 kHz. *Press ▶ below the frame (Adobe Acrobat Reader / Okular); other viewers show the final frame. Same animation as video `09_sampling_rate.mp4`.*

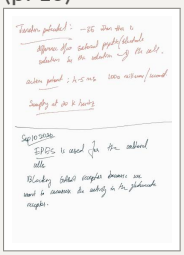
PART II

Sep 10, 2026 — Cultured neurons, astrocytes and synaptic currents

This part follows notebook pages 10–22: recordings from cultured neurons, including co-cultures with astrocytes, plus the pharmacology and plasticity discussed that day.

15 EPSCs in cultured cells, and why GABA receptors are blocked

From your notes (p. 10)



Sep 10 2026. EPSCs is used for the cultured cells. Blocking GABA receptors because we want to measure the activity in the glutamate receptors.

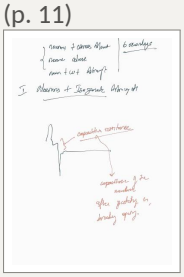
An **EPSC** (excitatory postsynaptic current) is the inward current through glutamate receptors, recorded in voltage clamp (usually at −70–−60 mV). In cultures, one usually records:

- **sEPSCs** (spontaneous): whatever happens in the network, including release triggered by presynaptic spikes.
- **mEPSCs** (miniature): recorded in TTX (0.5–1 µM) to block Na^+ channels and therefore spikes. Each event is one vesicle released spontaneously. **Frequency** reflects the number of synapses × release probability (mostly presynaptic). **Amplitude** reflects how many receptors are under each release site (postsynaptic).

Why block $GABA_A$? Inhibitory currents overlap with and contaminate excitatory events. Blocking them leaves a clean glutamatergic signal. Common blockers: **picrotoxin** (50–100 µM, pore blocker), **bicuculline** (10–20 µM, competitive antagonist), **gabazine / SR-95531** (5–10 µM). An alternative is to separate EPSCs and IPSCs by holding potential (Section 34).

16 Experimental groups: neurons with CANVAS, isogenic or WT astrocytes

From your notes (p. 11)



{ neurons + CANVAS astrocytes; neurons alone; neurons + WT astrocytes } | 6 [coverslips / conditions?]. I. Neurons + isogenic astrocytes.

CANVAS is *Cerebellar Ataxia, Neuropathy and Vestibular Areflexia Syndrome*, a late-onset ataxia caused by a biallelic intronic AAGGG repeat expansion in *RFC1*. Like SCA1, it is a repeat-expansion ataxia with prominent cerebellar involvement.

Why co-culture neurons with astrocytes? Astrocytes are not passive packing material (though electrically they look passive; Section 6). They:

- remove glutamate from synapses (transporters EAAT1/GLAST and EAAT2/GLT-1), which shapes EPSC decay and prevents excitotoxicity;
- buffer extracellular K^+ (Kir4.1), which keeps resting potentials and excitability stable;
- secrete synapse-promoting factors (e.g. thrombospondins, glypicans), which changes synapse *number*.

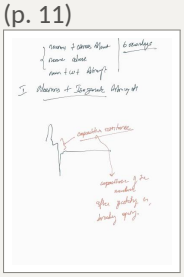
What each group lets you conclude.

Neurons alone	baseline: what the neurons do without astrocyte support
+ WT astrocytes	the effect of healthy astrocytes (usually more synapses, larger/more frequent EPSCs)
+ CANVAS astrocytes	does disease in astrocytes alone change neuronal synapses or excitability?
+ isogenic astrocytes	the *key control*: same genetic background as the CANVAS line but with the expansion corrected, so any difference is attributable to the mutation, not to donor-to-donor variation

Useful readouts: mEPSC/sEPSC frequency and amplitude, EPSC decay time (glutamate clearance), passive properties (capacitance and input resistance, next section), and intrinsic excitability (F–I curves, Section 37).

17 The capacitive transient after break-in

From your notes (p. 11)



[Trace with a sharp spike at the step.] Capacitive [resistance?] → capacitance of the membrane after patching in, breaking open.

The sharp spike at the edge of each seal-test pulse, after break-in, is the **capacitive transient**. The membrane is a thin insulator separating two conductors (cytoplasm and bath): a capacitor. Its capacitance is about 1 µF/cm^2, so **C_m measures cell size**. Changing the voltage across a capacitor requires moving charge ($Q = C\,\Delta V$). That charge has to flow through the pipette's access resistance, which produces a brief current spike that decays exponentially.

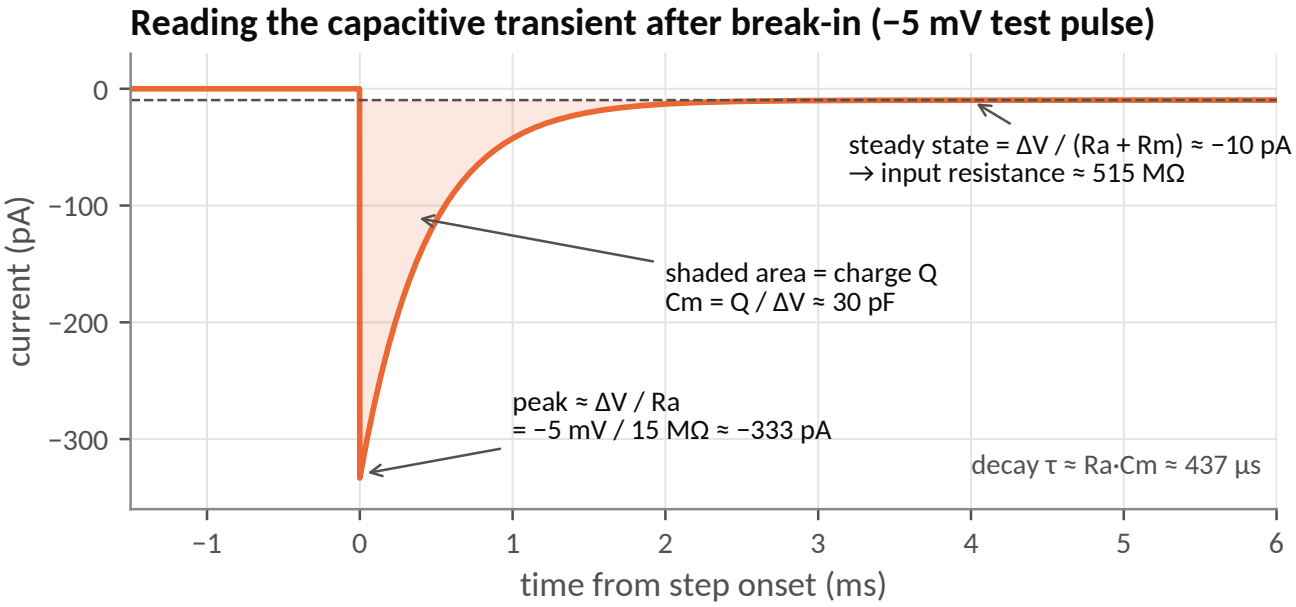
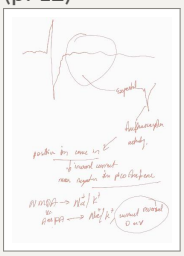


Figure 17: How to read the three numbers every recording should report from the membrane test.

Parameter	Read from	Why it matters / typical QC
R_a (access / series)	peak: $\Delta V / I_{peak}$	resistance between amplifier and cell interior. Voltage errors ($I \times R_a$) and slow clamp. Aim $<$ 20–25 MΩ and stable (<20% change); discard otherwise
R_m / R_{in} (input)	steady state: $\Delta V / I_{ss}$	how "leaky" the cell is; high in small or immature cells (hundreds of MΩ–GΩ)
C_m (capacitance)	area under transient: $Q/\Delta V$ (or τ / R_a)	cell size; normalise currents as pA/pF to compare cells
Holding current	baseline at −70 mV	a big leak ($<$ −100–−200 pA) means a poor seal or unhealthy cell

18 Expected AMPA receptor activity; AMPA vs NMDA

From your notes (p. 12)



[Trace with break-in transient, then a flat baseline:] Expected → AMPA receptor activity. Positive ions come in → inward current → means negative in picoampere. NMDA → Na^+/K^+ vs. AMPA → Na^+/K^+: current reversal 0 mV.

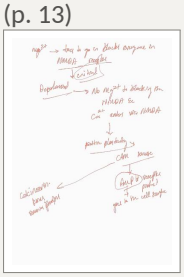
Both receptor types are non-selective cation channels, so both reverse near 0 mV. They differ in **speed** and **voltage dependence**:

	AMPA	NMDA
Kinetics	rise <1 ms, decay 2–10 ms	rise 5–10 ms, decay 50–300 ms
Ions	Na^+, K^+ (Ca^{2+} only if GluA2-lacking)	Na^+, K^+, **Ca^{2+}**
Voltage dependence	none (linear I–V)	Mg^{2+} block at negative voltages
Needs	glutamate	glutamate **+** co-agonist (glycine/D-serine) **+** depolarization
Blockers	NBQX, CNQX (10–20 µM)	D-APV / AP5 (50 µM)

AMPA/NMDA ratio. Because of the Mg^{2+} block, at −70 mV the EPSC is almost pure AMPA. At 40 mV the block is relieved and the slow NMDA component dominates (usually measured 50 ms after the stimulus, when AMPA has decayed). Their ratio is a standard synaptic readout, and it needs a Cs^+ internal to hold at 40 mV (Section 34).

19 The Mg^{2+} block, and how calcium becomes plasticity

From your notes (p. 13)



Mg^{2+} → tries to go in, blocks everyone in NMDA receptor (critical). Depolarized → no Mg^{2+} blocking the NMDA R. Ca^{2+} enters via NMDA → positive plasticity → CAM kinase → AMPA receptor product goes to the cell surface. Calcineurin: kinase, removes phosph.

At negative voltages, extracellular Mg^{2+} is driven into the NMDA pore by the electric field and gets stuck there, blocking it. Depolarization pushes it back out. Glutamate opens the gate, but current only flows if the cell is also depolarized. That makes the NMDA receptor a **molecular coincidence detector**: presynaptic activity (glutamate) *and* postsynaptic activity (depolarization) must happen together. This is the cellular version of Hebb's rule, "cells that fire together wire together".

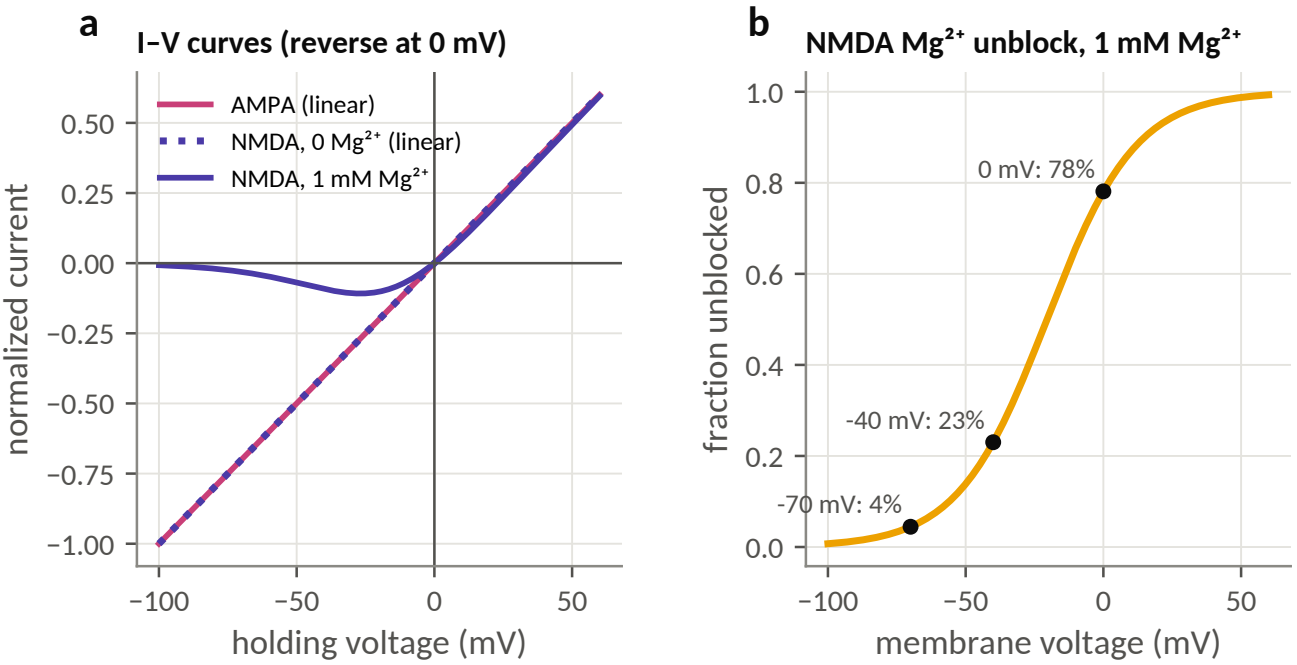


Figure 18: **a**, AMPA currents are linear in voltage. NMDA currents in 1 mM Mg^{2+} are tiny at negative voltages and appear on depolarization: the "J-shaped" I–V. **b**, Fraction of NMDA receptors not blocked, from the Jahr–Stevens equation $g(V) = 1/(1 + [Mg^{2+}]/3.57\,\text{mM} \cdot e^{-0.062V})$.

From Ca^{2+} to a stronger synapse (LTP). Ca^{2+} entering through NMDA receptors binds **calmodulin**. Ca^{2+}/calmodulin activates **CaMKII**, which:

1. autophosphorylates (Thr286), so it stays active after Ca^{2+} falls: a molecular memory;

2. phosphorylates AMPA receptors (GluA1 Ser831), increasing their conductance;
3. promotes insertion of more AMPA receptors into the synapse (your "goes to the cell surface").

The same glutamate release then produces a bigger EPSC.

A gentle correction **Calcineurin is a phosphatase, not a kinase.** Kinases *add* phosphate groups; phosphatases *remove* them. Your note's meaning ("removes phosph") is right, only the label is off. Calcineurin (PP2B) is activated by *moderate*, prolonged Ca^{2+} rises. It dephosphorylates AMPA receptors (GluA1 Ser845) and activates PP1, which leads to AMPA receptor removal (endocytosis) and **LTD**.

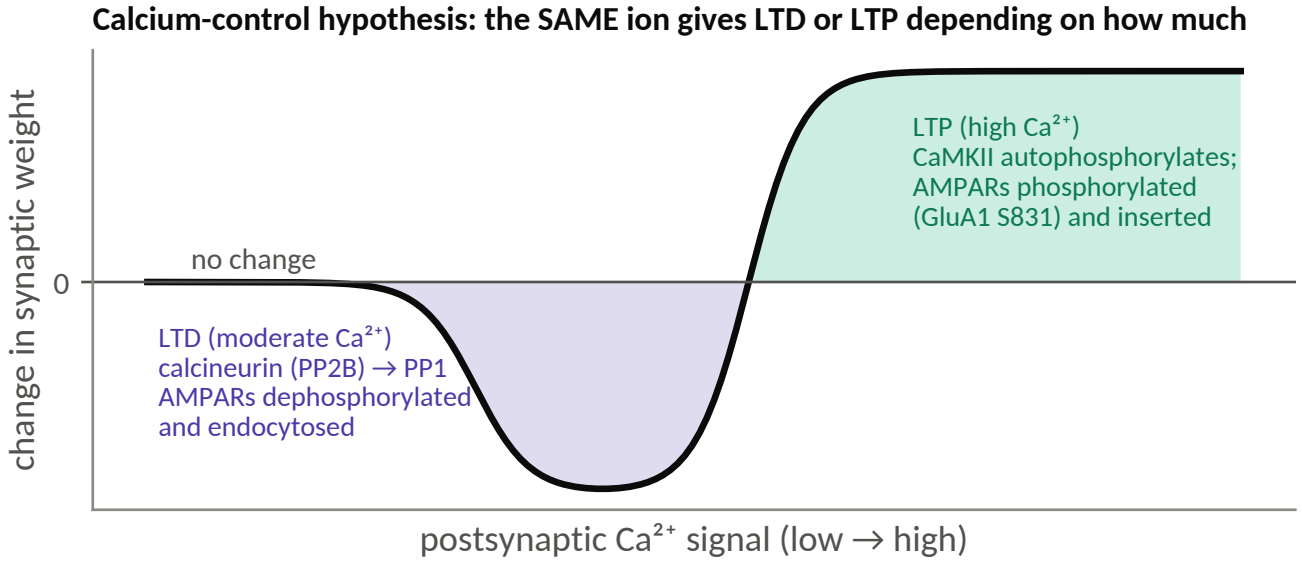


Figure 19: The calcium-control idea (Shouval, Bear & Cooper, 2002): the amount of postsynaptic Ca^{2+} decides the direction of plasticity. Calcineurin has higher Ca^{2+}/calmodulin affinity and wins at moderate levels; CaMKII needs high levels.

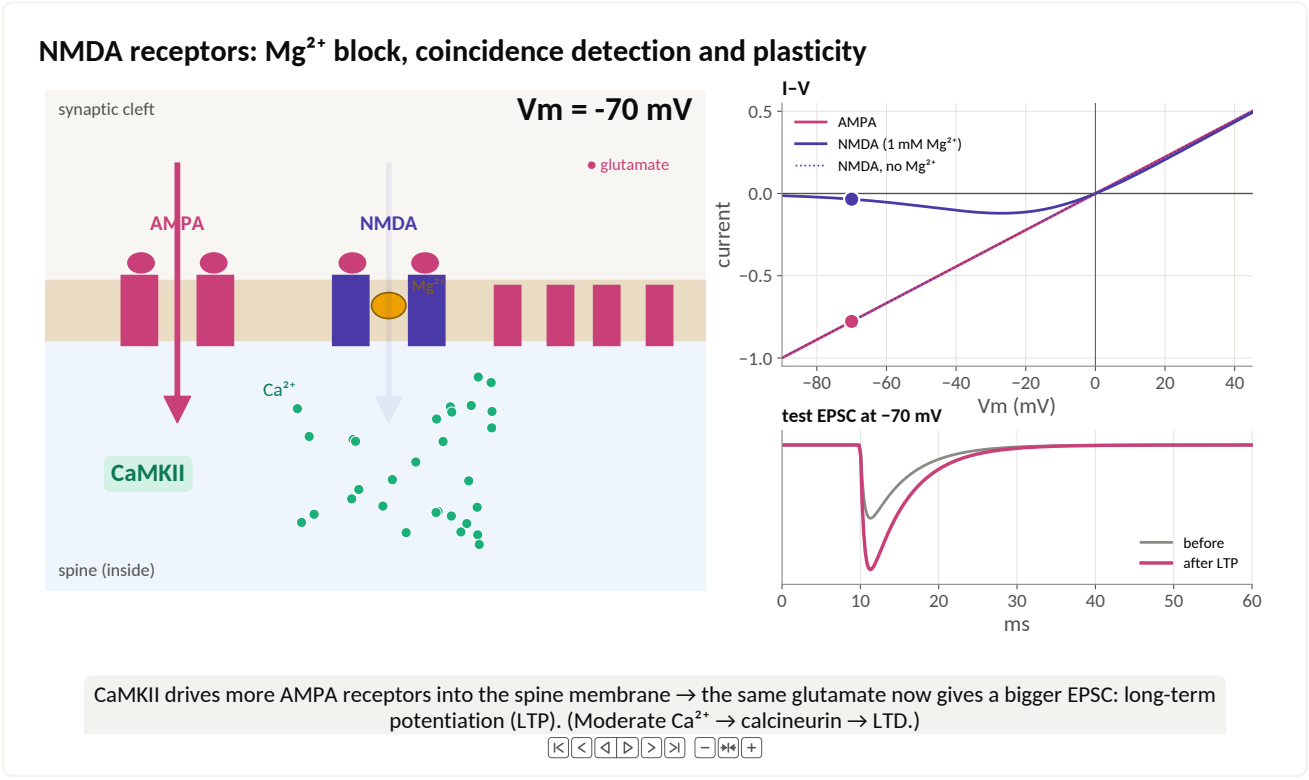
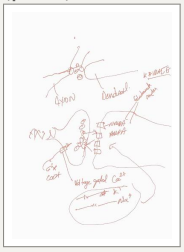


Figure 20: Animation — NMDA receptors, Mg^{2+} block and LTP. Voltage sweeps from −80 to 40 mV while glutamate is bound. The Mg^{2+} ion leaves the pore, Ca^{2+} enters, CaMKII switches on, and extra AMPA receptors are inserted, enlarging the EPSC. *Press ▶ below the frame (Adobe Acrobat Reader / Okular); other viewers show the final frame. Same animation as video* `07_nmda_mg_block_plasticity.mp4`.

20 Anatomy of a glutamatergic synapse

From your notes (p. 14)



[Drawing: axon, dendrite; presynaptic terminal with vesicles; NMDA, AMPA, KAINATE (glutamate receptors) on the dendrite; voltage-gated Ca^{2+} on the terminal; K^+, Na^+ channels.]

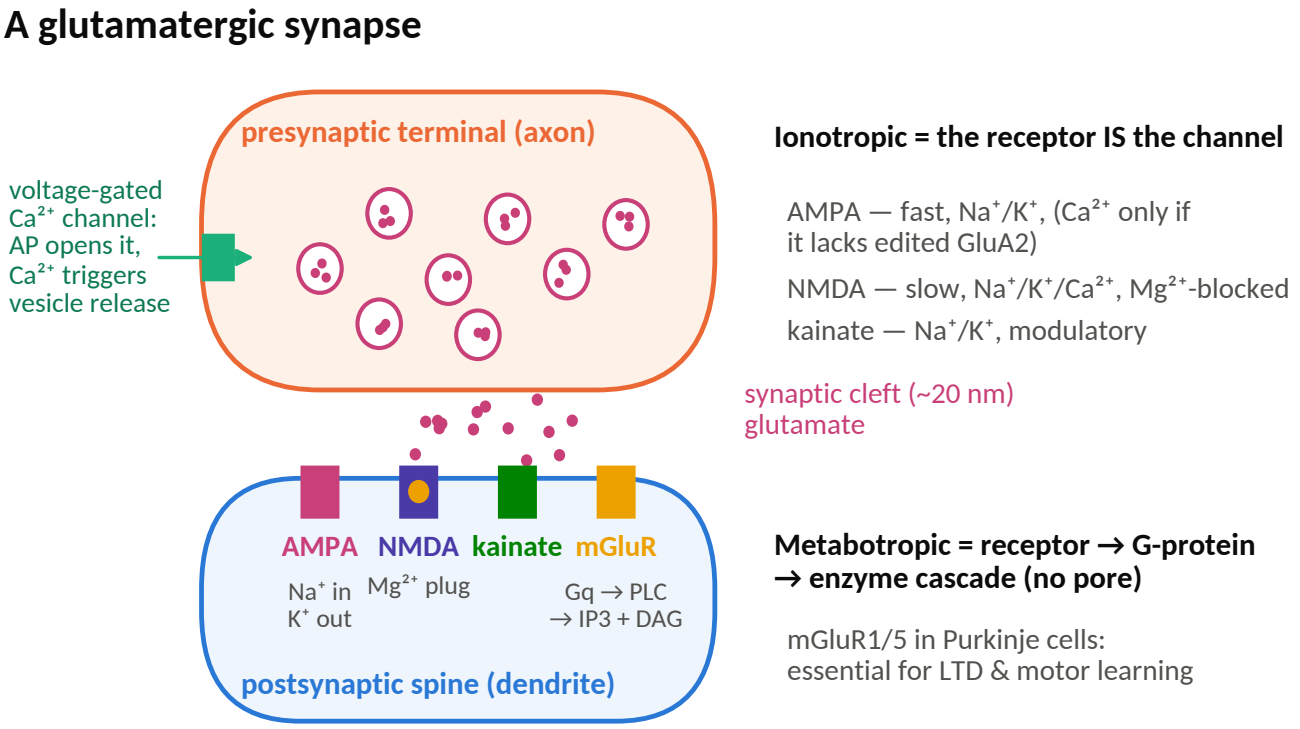
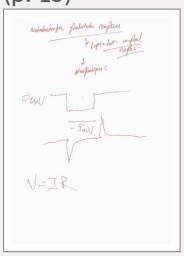


Figure 21: Your synapse drawing, redrawn. An action potential arriving at the terminal opens voltage-gated Ca^{2+} channels. Ca^{2+} entry triggers vesicle fusion within a fraction of a millisecond, and glutamate crosses the ~20 nm cleft to the receptors on the spine. Voltage-gated Na^{+} and K^{+} channels (bottom of your drawing) are what carry the action potential along the axon to the terminal.

21 Metabotropic glutamate receptors

From your notes (p. 15)



Metabotropic glutamate receptor → G-protein-coupled receptor → phospholipase C. [Seal-test sketch: 0 mV, step to –5 mV; current with transients.] V = IR.

mGluRs have no pore. Group I (mGluR1 and mGluR5) couple to **Gq → phospholipase C (PLC)**, which cleaves the membrane lipid PIP_2 into:

- **IP_3**, which opens IP_3 receptors on the endoplasmic reticulum → Ca^{2+} release from internal stores;
- **DAG**, which activates protein kinase C (PKC).

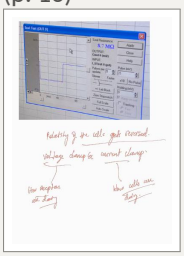
In Purkinje cells, mGluR1 at parallel-fibre synapses drives a slow EPSC (via TRPC3 channels) and is *required*

for PF-LTD and cerebellar motor learning. mGluR1-knockout mice are ataxic. IP_3R1 (*ITPR1*) and TRPC3 are themselves ataxia genes. This pathway is a recurring theme across the spinocerebellar ataxias.

The sketch on the same page is the seal test again: 0 mV holding, a −5 mV step, and the current response with transients. It is covered in Section 3 and Section 17.

22 The Seal Test window; what each mode tells you

From your notes (p. 16)



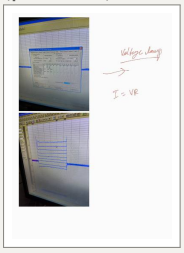
[Screenshot: Seal Test (OUT 0), seal resistance 8.7 MΩ, pulse −5 mV, holding 0 mV.] Polarity of the cells gets reversed. Voltage clamp → how receptors are doing. Current clamp → how cells are doing.

Reading the screenshot. Output "Cmd 0 (mV)" and input "I_STest 0 (pA)": the software is commanding voltage and reading current, so this is voltage clamp. "Pulse −5 mV", "Holding 0 mV", and a flat square of about −575 pA give 8.7 MΩ. That is pipette resistance in the bath: no seal yet.

"Polarity gets reversed" most likely refers to the sign convention that trips everyone up. A *depolarizing* event (positive ions entering) shows up as a *negative, downward* current in voltage clamp, but as a *positive, upward* voltage change in current clamp. The same biology appears with opposite polarity in the two modes (Animation 6).

23 Voltage-clamp step protocol and the I–V family

From your notes (p. 17–18)



[Screenshots: Edit Protocol "CS_VC_IV_-80-20_hold_-60" [name?]; a family of current traces.] Voltage clamp → I = VR. [p. 18: a fast inward spike at step onset:] Na^+ ion spike. Increasing current triggers increasing intake of Na^+. Silver chloride. Injected current vs voltage-gated current. −ve current is positive ions moving inside. NMDA receptors.

The protocol. The file name reads as *Cs internal, voltage clamp, I–V, steps from −80 to 20 mV, holding −60 mV*. The cell is held at −60 mV, stepped to each test voltage in 10 mV increments, and returned. For every step you record the current the amplifier has to inject.

A gentle correction Ohm's law is **I = V / R** (current = voltage divided by resistance), not I = VR. In conductance form, **I = g·V** with $g = 1/R$. Your earlier V = IR entries are correct.

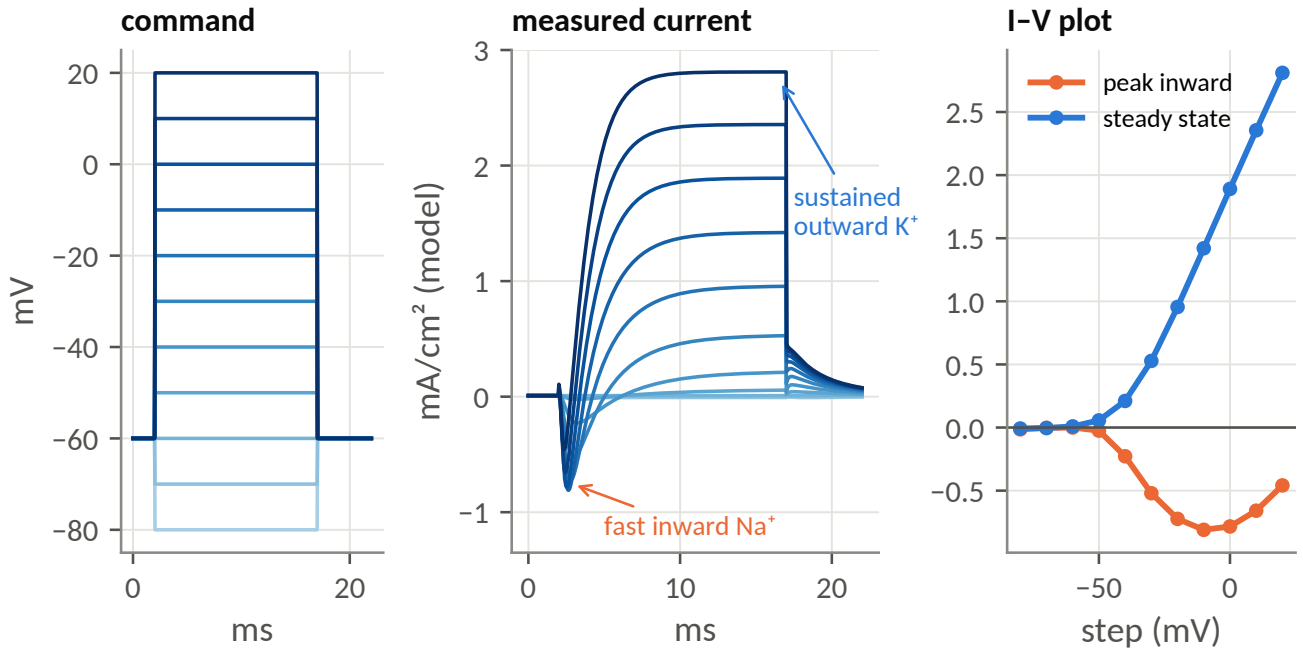


Figure 22: A step family and its I–V plot (Hodgkin–Huxley model, K^+ internal). Small steps give only leak current. Past about −50 mV a fast inward Na^+ current appears and grows with depolarization; larger steps also recruit a slow, sustained outward K^+ current.

"Na^+ ion spike": Na^+ current or capacitive transient? Both appear as fast spikes at the step onset. Tell them apart like this:

- A **capacitive transient** appears at *every* step, including hyperpolarizing ones. Its size scales linearly with step size, and it has a mirror-image partner at step *offset*. It is instantaneous (tens to hundreds of µs).
- A **Na^+ current** appears only for steps above about −50 mV. It has a finite rise (0.3–1 ms), is always inward, peaks around −20–0 mV and then shrinks, has no partner at offset, and is abolished by TTX.

Your observation, "increasing current triggers increasing intake of Na^+", is the left half of the Na^+ I–V. Bigger steps open more Na^+ channels. Very large steps reduce the Na^+ current again as V_m approaches E_{Na}.

Injected current vs voltage-gated current. The amplifier's injected current is the mirror image of the current through the cell's own channels. In VC, what you plot as "current" is the channel current, with the inward-is-negative convention.

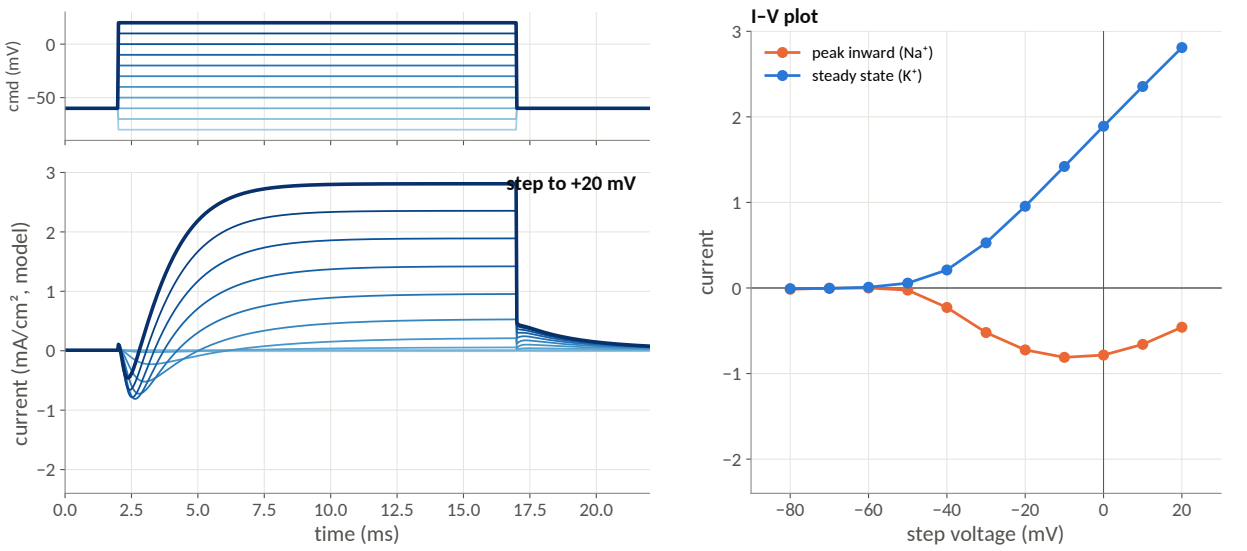
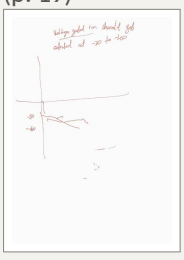


Figure 23: Animation — Voltage-clamp I–V. Each sweep is drawn in turn, and its peak (Na^+) and steady-state (K^+) currents are added to the I–V plot. *Press ▶ below the frame (Adobe Acrobat Reader / Okular); other viewers show the final frame. Same animation as video* `05_voltage_clamp_iv.mp4`.

24 Voltage-gated channels activate around −40 to −30 mV

From your notes (p. 19)



Voltage-gated ion channels get activated at −30 to −40. [Sketch: currents appearing between −30 and −60.]

Each voltage-gated channel has an **activation curve**: the fraction of channels open at steady state as a function of voltage, usually a sigmoid (Boltzmann). "Activation threshold" is where that curve lifts off the floor. Na_V channels begin opening around −55–−45 mV and are half-activated around −30–−20 mV. Many K_V channels start around −40–−30 mV. The exact numbers depend on subtype (Nav1.1/1.2/1.6, Kv1/3/4…), temperature and the cell.

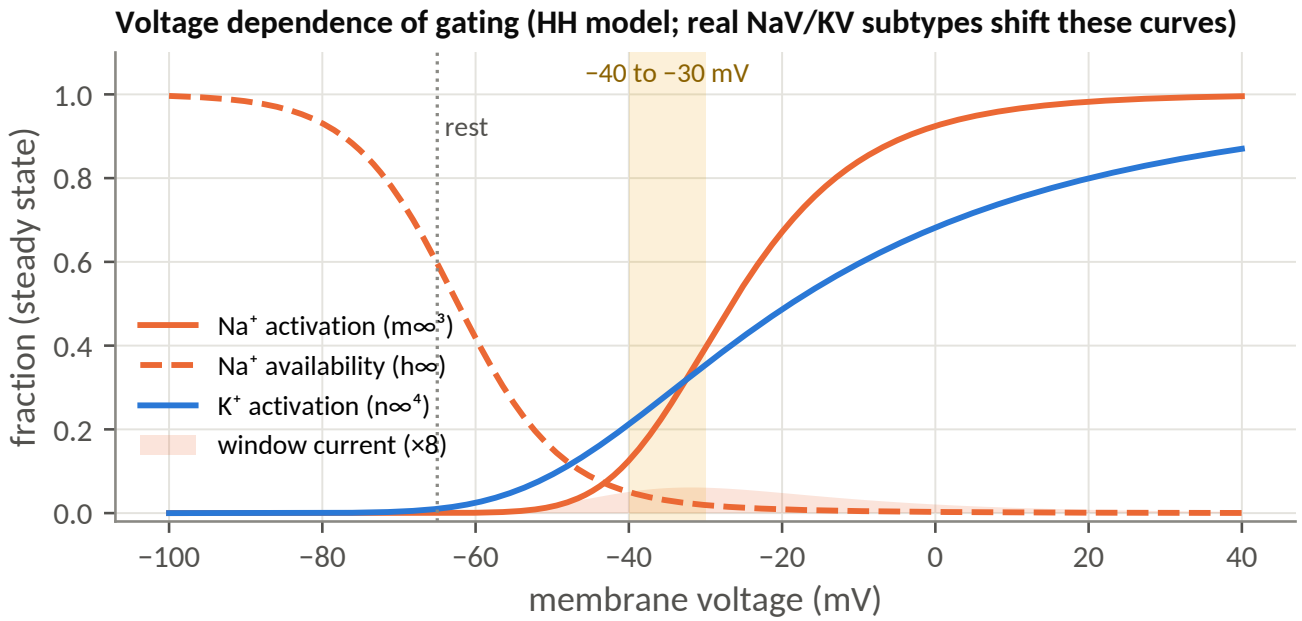
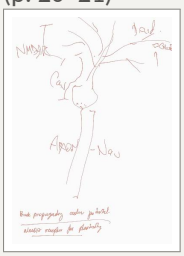


Figure 24: Steady-state gating vs voltage. Where Na^+ activation and availability overlap (shaded) there is a small persistent "window current". Real Na_V and K_V subtypes shift these curves by tens of mV.

25 Where each channel lives; back-propagating APs and plasticity

From your notes (p. 20–21)



[Neuron drawing: NMDAR and Ca_V on dendrites, glutamate at a spine, Na_V on the axon.] Back-propagating action potential. NMDA receptor for plasticity. Current → depolarize the cells/dendrites (AMPA, dendrite) → glutamate + AMPA → Na^+ & K^+ inward → dendritic spike → axon. Voltage-gated channels: Na^+, K^+, Ca^{2+} | receptor channels are different.

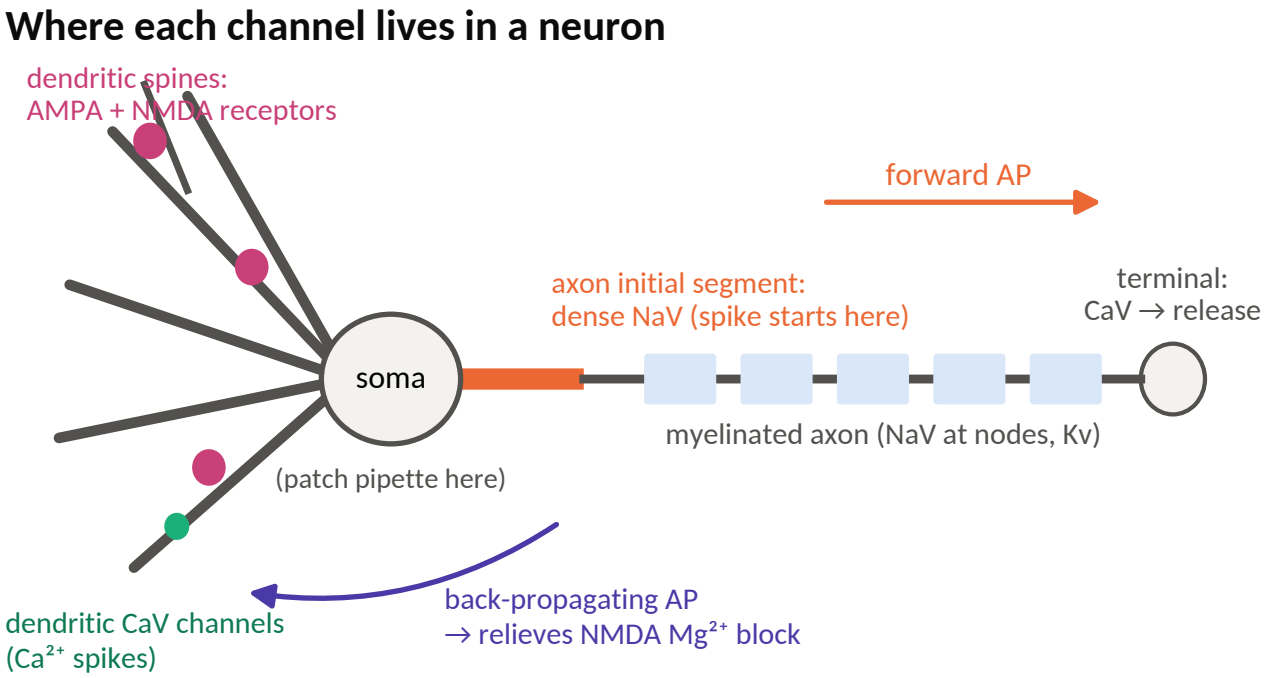


Figure 25: Channel geography. Receptor (ligand-gated) channels sit at synapses on dendritic spines. Voltage-gated Na^+ channels are concentrated at the axon initial segment, where spikes start, and at nodes. Ca^{2+} channels are on dendrites and at terminals.

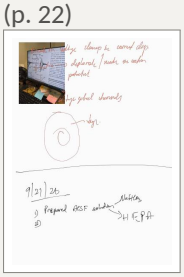
The flow in your notes, step by step:

1. Glutamate opens AMPA receptors → inward Na^+ current (Na^+ in exceeds K^+ out) → local depolarization of the spine and dendrite.
2. If many inputs coincide, dendritic voltage-gated channels (Ca^{2+}, sometimes Na^+) and NMDA receptors can produce a regenerative **dendritic spike**.
3. Depolarization spreads to the soma and the axon initial segment. If it crosses threshold there, an AP fires and travels down the axon.
4. The AP also travels *backwards* into the dendrites: the **back-propagating AP (bAP)**. At the spines, the bAP depolarization expels Mg^{2+} from NMDA receptors that recently bound glutamate. This is how the synapse "knows" that it helped make the cell fire, the basis of spike-timing-dependent plasticity.

Receptor channels vs voltage-gated channels. Receptor channels (AMPA, NMDA, $GABA_A$) open because a transmitter binds. Voltage-gated channels (Na_V, K_V, Ca_V) open because the voltage changes. The first set of channels decides *what input* the cell receives; the second decides *what the cell does with it* (Section 36).

26 Reading a paper figure: from receptors to voltage-gated channels

From your notes (p. 22)



[Photo of a paper on developmental changes in AMPA receptors / GluR2 in layer 5 pyramidal neurons, with I–V plots.] Breaks in voltage clamp & current clamp. Glutamate → depolarize / neuron or action potential → voltage-gated channels.

The paper on your screen studies how AMPA receptors change with development (GluR2 = GluA2). The I–V plots in such figures are how GluA2 content is diagnosed:

- **GluA2-containing** AMPARs: **linear** I–V, Ca^{2+}-impermeable.
- **GluA2-lacking** AMPARs: **inwardly rectifying** I–V (little outward current at positive voltages, because intracellular polyamines block them), Ca^{2+}-permeable.

The *rectification index* (current at 40 divided by current at −60 mV, scaled by driving force) quantifies this. It is measured in voltage clamp with spermine added to the internal. The chain in your note, "glutamate → depolarize → AP → voltage-gated channels", is the sequence you would test by switching from voltage clamp (receptors) to current clamp (cell output).

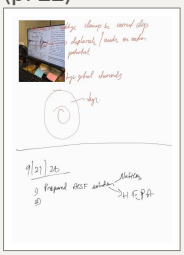
PART III

9/21/26 — aCSF, ion gradients, plasticity, cell-attached and internals

This part follows notebook pages 22–31.

27 Preparing aCSF: bicarbonate vs HEPES

From your notes (p. 22)



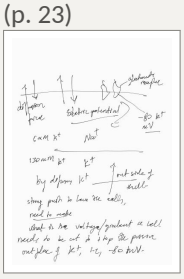
9/21/26. 1) Prepared aCSF solution → $NaHCO_3$, HEPES. 2) …

Two buffer systems appear in slice work:

- **Bicarbonate ($NaHCO_3$)** is physiological, and HCO_3^- also passes through $GABA_A$ channels. It *requires* continuous 5% CO_2 (Section 1.4).
- **HEPES** is a synthetic buffer (pK_a ~7.5) that holds pH without CO_2. It is used in internal (pipette) solutions and in NMDG/HEPES cutting and holding solutions, often *added* to bicarbonate aCSF (5–20 mM) as extra protection in case bubbling falters. It is also used in recording solutions for cultures outside the incubator.

28 Diffusion force vs electrical force: the Nernst potential

From your notes (p. 23)



[Membrane with arrows:] diffusion force, electric potential; glutamate receptor; –80 mV K^+. 5 mM K^+ [outside], 130 mM K^+ [inside]. By diffusion K^+ [goes] outside of the cell, strong push to leave the cells. Need to make: what is the voltage/gradient a cell needs to be at to stop the passive outflux of K^+, i.e. –80 mV.

This is the most important calculation in electrophysiology, and your note states the question exactly right. Two

forces act on each ion:

1. **Diffusion (chemical) force**: ions move from high to low concentration. For K^+ (130 mM in, 5 mM out) that means outward.
2. **Electrical force**: charges move in the electric field. A negative interior pulls K^+ back in.

The voltage at which they balance, so that there is *no net flow*, is the ion's equilibrium (Nernst) potential:

$$E_{\text{ion}} = \frac{RT}{zF} \ln \frac{[\text{ion}]_{\text{out}}}{[\text{ion}]_{\text{in}}} \approx \frac{61.5\ \text{mV}}{z} \log_{10} \frac{[\text{ion}]_{\text{out}}}{[\text{ion}]_{\text{in}}} \quad (37\,^\circ\text{C})$$

For K^+: $E_K = 61.5 \log_{10}(5/130) = 61.5 \times (-1.415) = -$**87 mV**. That is your "−80 mV" to within rounding.

Ion	z	Outside (mM)	Inside (mM)	E at 37 °C	E at 25 °C
K^+	+1	5	130	−87 mV	−84 mV
Na^+	+1	140	12	66 mV	63 mV
Na^+ (your values)	+1	140	5	89 mV	86 mV
Cl^-	−1	130	10	−69 mV	−66 mV
Ca^{2+}	+2	2	0.0001	132 mV	127 mV

Why the resting potential is about −65 mV, not −87 mV. At rest the membrane is mostly, but not only, permeable to K^+. The small Na^+ leak pulls V_m a little toward E_{Na}. The Goldman–Hodgkin–Katz equation weights each ion by its permeability:

$$V_m = 61.5 \log_{10} \frac{P_K[K]_o + P_{Na}[Na]_o + P_{Cl}[Cl]_i}{P_K[K]_i + P_{Na}[Na]_i + P_{Cl}[Cl]_o}$$

With $P_K : P_{Na} : P_{Cl} = 1 : 0.05 : 0.45$ this gives −65 mV.

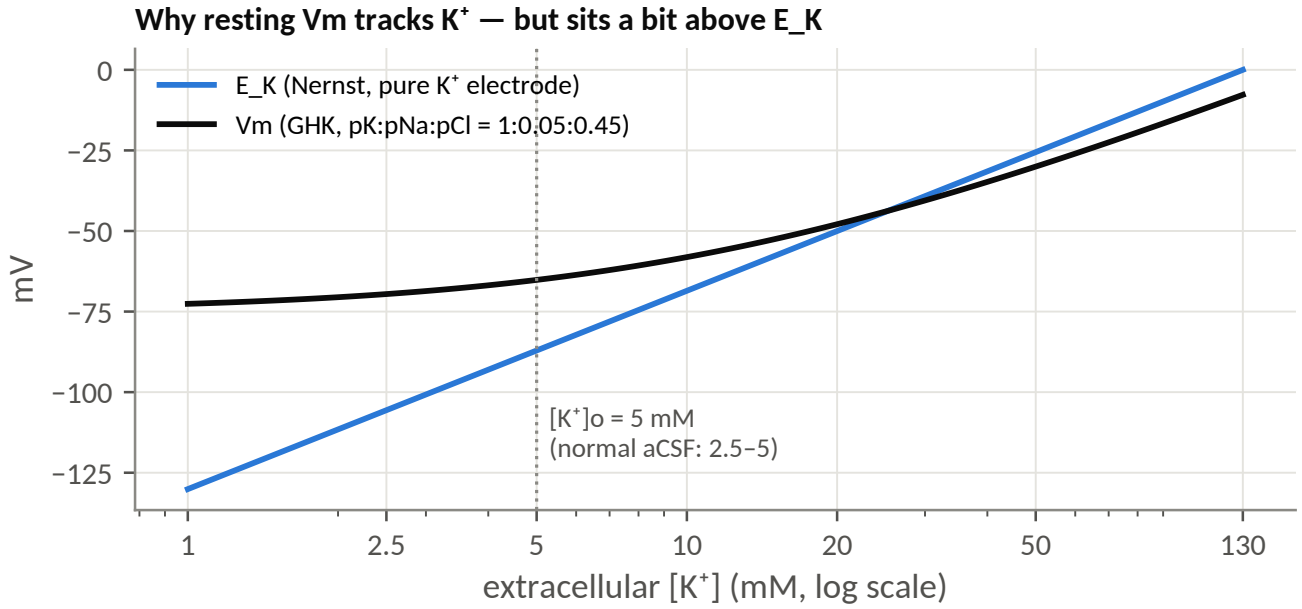


Figure 26: Change extracellular K^+ and V_m follows E_K. This is why aCSF K^+ (2.5–5 mM) matters so much, and why raising bath K^+ is a classic way to depolarize neurons.

How many ions actually move? Surprisingly few. Charging $1\,\mu F/cm^2$ of membrane by 87 mV needs about 10^{-12} mol of K^+ per cm^2, a tiny fraction (~1 in 10^5) of the K^+ inside a typical cell. Concentrations stay essentially unchanged. The Na^+/K^+-ATPase maintains the gradients over the long term.

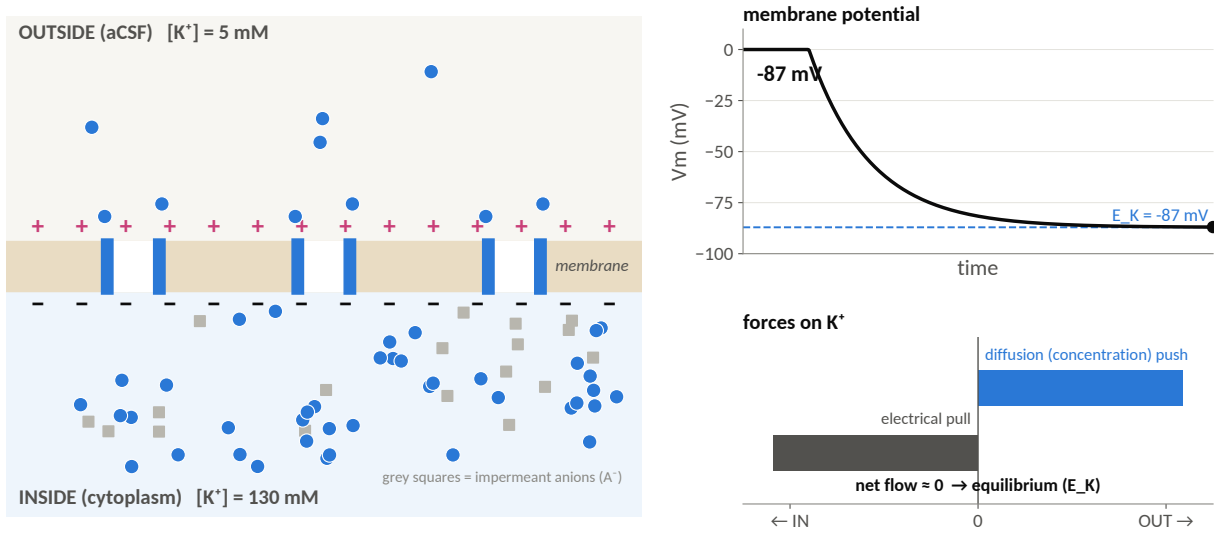


Figure 27: Animation — Where the resting potential comes from. K^+ ions leak out through open channels. Surface charge builds up, the voltage approaches E_K, and the electrical pull grows until it equals the diffusion push. *Press ▶ below the frame (Adobe Acrobat Reader / Okular); other viewers show the final frame. Same animation as video* `01_resting_potential.mp4`.

29 Sodium: how positive to keep Na^+ outside?

From your notes (p. 24)



140 mM Na^+ [outside]; 5 mM Na^+ [inside]. What is the minimum positivity of the cells to keep Na^+ outside: +40 mV.

A gentle correction The Nernst equation with your numbers gives $E_{\mathrm{Na}} = 61.5 \log_{10}(140/5) \approx +\mathbf{89\ mV}$. With a more typical intracellular Na^+ of 10–15 mM it is about 55–65 mV. The 40 mV is the typical *peak of the action potential*, and the spike peaks *below* E_{Na}. The reason: before V_m can get there, Na^+ channels inactivate and K^+ channels open (Fig. 9). So at 40 mV Na^+ would still flow *in* through any open channel; the inside must reach E_{Na} to stop it.

30 Slicing solutions: protecting neurons during the cut

From your notes (p. 24)



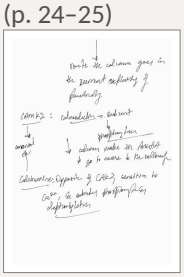
Warm slicing: add drugs to block glutamate receptors → [neuroprotection?]. Change → ions & get rid of calcium during slicing phase.

Cutting slices damages cells. Damaged cells dump glutamate and K^+, which depolarize their neighbours, open NMDA receptors, flood them with Ca^{2+}, and kill them (excitotoxicity). Protective strategies:

- **Low Ca^{2+}, high Mg^{2+}** cutting solution (e.g. 0.5–1 mM Ca^{2+}, 3–7 mM Mg^{2+}): less transmitter release, stronger NMDA block. This is your "get rid of calcium during slicing".
- **Replace Na^+** with sucrose, choline or **NMDG**. Less Na^+ influx means less swelling and fewer firing damaged cells.
- **Glutamate receptor blockers** during cutting/recovery (e.g. kynurenic acid, a broad-spectrum antagonist, or D-APV). This is your "add drugs to block glutamate receptors".
- **Temperature.** The classical approach is ice-cold cutting. For adult tissue, including mature cerebellum, cutting at physiological temperature can give healthier slices. See *"Physiological temperature during brain slicing enhances the quality of acute slice preparations"* (Huang & Uusisaari, 2013, *Front. Cell. Neurosci.*), and the follow-up protocol *"Slice it hot"*. This is likely what "warm slicing" refers to.

31 NMDA, AMPA and the route calcium takes

From your notes (p. 24–25)



NMDA & glutamate, AMPA → calcium + Na^+, + K^+ → route the calcium goes in, the recurrent complexity of plasticity. CAMK2: calmodulin → subunit → phosphorylates; calcium make the AMPA to go to more to the cell surface. Canonical. Calcineurin: opposite of CAMK2, sensitive to Ca^{2+}, activates phosphatases, dephosphorylation.

This restates and sharpens Section 19. It is correct: calcineurin is the Ca^{2+}-sensitive counterpart of CaMKII and drives dephosphorylation (see the correction box there). Two extra points about the *route* of Ca^{2+}:

- **Where Ca^{2+} comes from shapes what it does.** Ca^{2+} through NMDA receptors lands right next to CaMKII anchored at the postsynaptic density. Ca^{2+} through voltage-gated Ca^{2+} channels, GluA2-lacking AMPARs, or IP_3-receptor stores (mGluR1) produces different spatial and temporal signals and can drive different plasticity. In Purkinje cells, for example, climbing-fibre-evoked Ca^{2+} through voltage-gated channels drives PF-LTD.
- **"Recurrent complexity"**: CaMKII autophosphorylation, AMPA receptor trafficking, and Ca^{2+}-dependent feedback onto the channels themselves make plasticity history-dependent (metaplasticity).

32 Cell-attached recording

From your notes (p. 26)



*[Harp over slice, tissue.] Cell-attached recording: bring the pipette tip near the cells, to check the spiking. * More than one cell is usually recorded — cell signal sorting is done to separate the signal. Usually 15–20 hertz; discard cells spike like 10 Hz.*

In **cell-attached** mode the pipette touches or seals onto the membrane but does *not* break in. The cell's interior is untouched: no dialysis, no washout of second messengers. That makes it the least invasive way to measure **spontaneous firing rate and regularity**. Each spike appears as a small biphasic current (tens of pA) through the patch.

- **Tight-seal** (GΩ) cell-attached sees only the patched cell.
- **Loose-seal / loose-patch** (tens of MΩ) is gentler and can be repeated on many cells, but can also pick up **neighbouring cells**. Their spikes have different amplitudes and shapes, so they are separated by **spike sorting** (your "cell signal sorting").
- Record in voltage clamp with the command set so no net current flows (often 0 mV), or in current clamp at

I = 0, so you don't depolarize the cell.

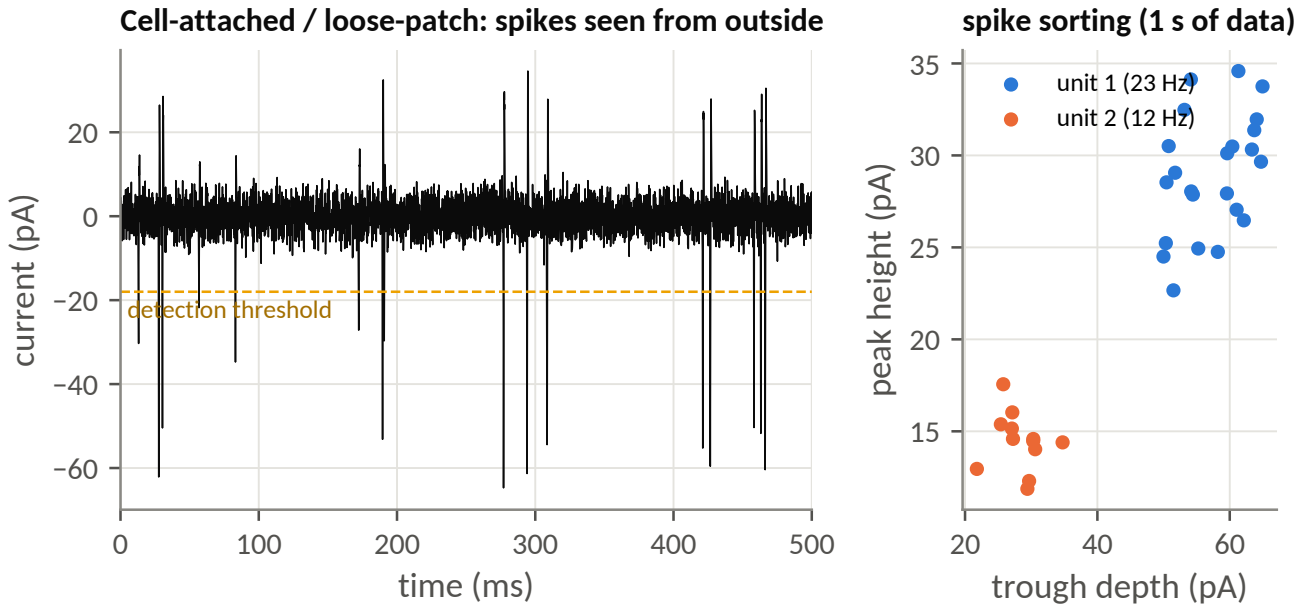


Figure 28: Simulated loose-patch recording with two units. Detection by threshold, then sorting by waveform features (here trough depth vs peak height), separates the cells.

Firing-rate criteria depend on cell type

The "usually 15–20 Hz, discard cells spiking at ~10 Hz" rule is a health or identity criterion for the cells in your prep. Purkinje cells in healthy acute slices typically fire spontaneously at tens of Hz, and reduced or irregular firing is itself an early SCA1 phenotype. So be careful: a "discard slow cells" rule could remove exactly the cells that carry the disease signal. Worth writing the rule's rationale into the protocol.

33 Ionotropic vs metabotropic; the transmitter systems

From your notes (p. 27)



Metabotropic, ionotropic → glutamate, GABA → GABA is inhibitory, chloride. Serotonin → ascending system → long axons; acetylcholine. High [Cl^-] outside, low chloride inside. −70 reverse potential, similar to Na^+. GABA → inhibition & sensory perception.

- **Fast point-to-point transmitters**: glutamate (excitatory) and GABA (inhibitory; glycine in brainstem and spinal cord). Each has ionotropic receptors (AMPA/NMDA/kainate; $GABA_A$) and metabotropic receptors (mGluRs; $GABA_B$, which opens K^+ channels via $G_{\beta\gamma}$).

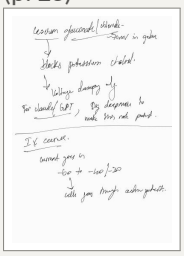
- **Neuromodulators**: serotonin, acetylcholine, noradrenaline, dopamine. Small nuclei in the brainstem and basal forebrain send **long, widely branching ascending axons** across the brain (your "ascending system"). They act mostly through metabotropic receptors to change the *gain and state* of circuits rather than carry specific messages.

Chloride. Like Na^+, Cl^- is high outside and low inside ("similar to Na^+"). Because it is negatively charged, its equilibrium potential is negative: $E_{Cl} \approx -70$ mV. Opening $GABA_A$ channels therefore pulls V_m toward −70 mV. Below threshold, that is *inhibition*, whether V_m moves slightly up or down. Even when V_m barely changes, the open channels "short-circuit" excitatory input (**shunting inhibition**). This sharpening of responses is part of why inhibition matters for sensory perception.

Important for cultured neurons Low intracellular Cl^- is not automatic. Mature neurons export Cl^- with the transporter KCC2. Immature neurons, including many iPSC-derived neurons in culture, express mostly NKCC1, which imports Cl^-. In those cells E_{Cl} can sit around −50–−40 mV, so GABA is **depolarizing** and can even be excitatory. Keep this in mind when interpreting GABA-related results in your co-cultures.

34 Deep dive: cesium-based internal solutions

From your notes (p. 28)



Cesium gluconate / chloride. –70 mV in GABA. → Blocks potassium channels → voltage clamping only. ***For Claude/GPT: dig deeper to make this note point.***

34.1 What the internal solution is and why it matters

Once you break in, the pipette solution (the *internal*) replaces the cytoplasm near the soma within minutes. Its composition therefore *sets* the intracellular ion concentrations, which set the reversal potentials, which determine the direction and size of every current you record. Choosing an internal is choosing what your experiment can see.

	K-gluconate	**Cs-gluconate / Cs-methanesulfonate**	**CsCl (high Cl^-)**
Main cation	K^+ (physiological)	Cs^+ (blocks K^+ channels)	Cs^+
Internal Cl^-	low (~5–10 mM)	low (~5–10 mM)	high (~130 mM)
E_{Cl}	~−70 mV	~−70 mV	~0 mV
Current clamp	**yes**: APs, firing, F–I	**no**: spikes broad, cells depolarized	no
Best for	excitability, EPSPs, AP shape	EPSCs vs IPSCs by holding potential; AMPA/NMDA ratio; holding at 0 or 40 mV	large inward mIPSCs at −70 mV (with glutamate blockers)

34.2 Why cesium? Three reasons

1. **It blocks K^+ channels from the inside.** Cs^+ enters K_V, K_{ir}, K2P and GIRK pores but permeates very poorly, so it plugs them. Without K^+ currents contaminating the trace, what remains at depolarized potentials is synaptic current and non-K^+ channels. Tetraethylammonium (TEA, ~5–10 mM) is often added to block more thoroughly.
2. **Better space clamp.** Blocking resting K^+ conductance raises membrane resistance. Current then spreads further along the dendrites before leaking out (a longer length constant, $\lambda \propto \sqrt{R_m}$). Your somatic clamp therefore controls voltage further out into the dendrites, where most synapses are. With K^+ internal, distal synapses are poorly clamped and their currents are distorted.
3. **You can hold at depolarized voltages.** To isolate IPSCs at 0 mV, or record NMDA currents at 40 mV, you need the cell to survive there. With a K^+ internal, huge outward K^+ currents would flow and the clamp would fail. With Cs^+, the cell sits at 0 mV quietly.

34.3 Gluconate vs chloride: setting E_{Cl}

The anion decides the GABA reversal potential. With **Cs-gluconate** (low Cl^-), $E_{Cl} = -61.5 \log_{10}(130/10) \approx -69$ mV. That is your "−70 mV in GABA". This gives a neat trick:

- Hold at **−70 mV** ($= E_{Cl}$): GABA currents have zero driving force and vanish. You see **only EPSCs** (inward, down).
- Hold at **0 to 10 mV** ($= E_{glu}$): glutamate currents vanish. You see **only IPSCs** (outward, up).

This is the pharmacology-free alternative to blocking GABA receptors (Section 15). You can record both E and I *in the same cell* and compute an E/I ratio. With **CsCl** (high Cl^-), $E_{Cl} \approx 0$ mV. GABA currents are then large and *inward* at −70 mV, which makes small inhibitory events easier to detect. Glutamate receptors must then be blocked (NBQX + APV) to tell them apart.

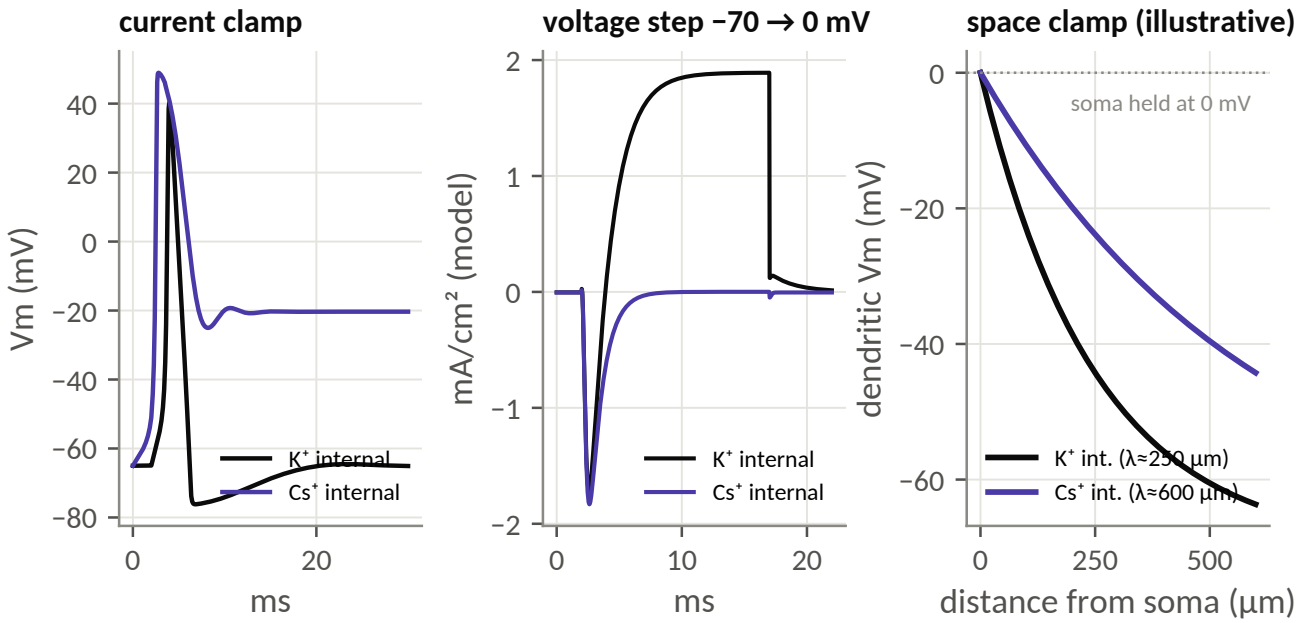


Figure 29: Why a Cs^+ internal is "voltage clamp only" (model). **Left**: in current clamp, with K^+ channels blocked, the spike cannot repolarize and the cell gets stuck depolarized. **Middle**: in voltage clamp, the outward K^+ current disappears and leaves the inward currents you want. **Right**: higher membrane resistance extends the length constant, so dendritic synapses are better clamped (illustrative λ values).

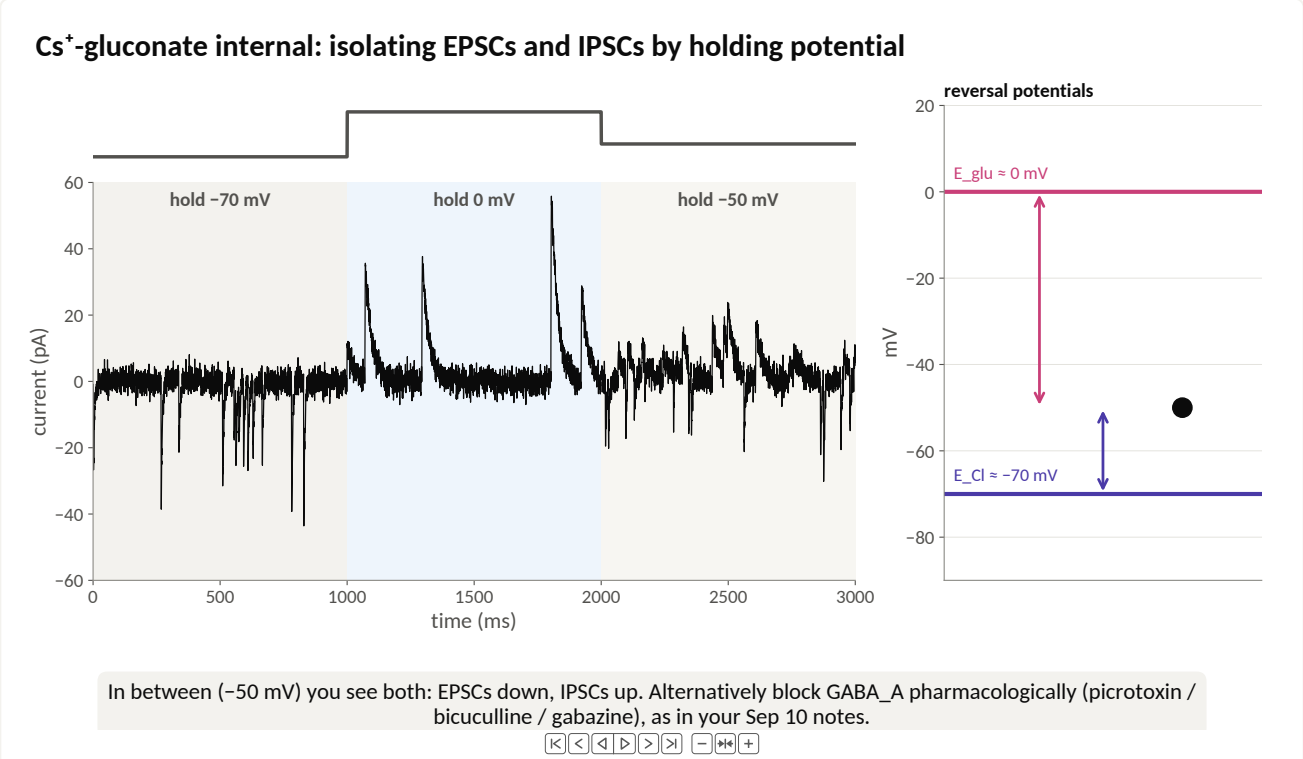


Figure 30: Animation — Isolating EPSCs and IPSCs with a Cs⁺ internal. Hold at −70 mV: only EPSCs. Hold at 0 mV: only IPSCs. At −50 mV: both. The right panel shows each receptor's driving force. *Press ▶ below the frame (Adobe Acrobat Reader / Okular); other viewers show the final frame. Same animation as video `08_isolating_epsc_ipsc.mp4`.*

34.4 Why "voltage clamping only"

With K^+ channels blocked, the membrane cannot repolarize after a spike. There is no normal AHP, and the resting potential drifts positive because the K^+ leak that set it is gone. Firing is therefore not physiological. Most Cs^+ internals also contain **QX-314** (~5 mM), a membrane-impermeant lidocaine derivative that blocks Na_V channels from inside. It prevents escaped spikes when you depolarize but abolishes APs entirely. Any intrinsic-excitability measurement (F–I curves, AP shape, rheobase) must therefore use a K^+-based internal, in separate cells.

34.5 A typical Cs-based internal (for orientation; use your lab's recipe)

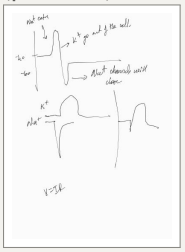
Component	mM	Purpose
Cs-methanesulfonate or Cs-gluconate	120–135	main salt; Cs^+ blocks K^+ channels
CsCl	5–10	sets low Cl^- ($E_{Cl} \approx -70$ mV)
HEPES	10	pH buffer (pH 7.2–7.3 with CsOH)
EGTA (or BAPTA)	0.2–10	Ca^{2+} buffer (high = blocks Ca^{2+}-dependent plasticity)
Mg-ATP / Na-GTP	4 / 0.3	energy, G-protein signalling
Phosphocreatine	10	ATP regeneration
QX-314	5	blocks Na_V (no escaped spikes)
TEA-Cl (optional)	5	extra K^+ channel block
Osmolarity ~285–295 mOsm, a little below the aCSF.		

34.6 Pitfalls and practical tips

- **Wait for dialysis.** Cs^+ and QX-314 take several minutes after break-in to reach the dendrites. Recording too early gives mixed K^+/Cs^+ behaviour.
- **Junction potential.** Gluconate internals have a sizeable LJP (~10–16 mV); methanesulfonate's is usually smaller. Correct holding potentials if exact reversal matters (Section 14). Your "−70 mV" might really be −80 mV at the membrane, so E_{Cl} nulls may appear slightly shifted.
- **Health at 0 mV.** Cells held depolarized for long periods load with Ca^{2+} and decline. Keep depolarized epochs short and watch R_a and holding current.
- **Space clamp is still imperfect**, especially in Purkinje cells with their enormous dendritic trees. Interpret the kinetics of distal events cautiously.
- **Cs^+ is not a universal blocker.** It is a weak blocker of some channels, and *extracellular* Cs^+ blocks HCN (I_h). Don't confuse internal and external cesium effects.

35 The inward current that starts the spike

From your notes (p. 28–29)



I[x?] current: current goes in −60 to −40/−30 → cells go through action potential. [p. 29:] Na^+ enters, K^+ go out of the cell, Na^+ channels will close. [K^+ current: outward hump; Na^+ current: sharp inward dip.] V = IR.

Between about −60 and −40–−30 mV, depolarization starts opening Na_V channels (Fig. 24). The resulting inward current (written I_{Na}; your note may say "I_x") depolarizes further and becomes regenerative. That is threshold. Your p. 29 drawing shows the classic Hodgkin–Huxley voltage-clamp result: Na^+ current is a **fast, transient, inward** dip (it self-terminates by inactivation), while K^+ current is a **delayed, sustained, outward** hump. In current clamp these two, one after the other, *are* the action potential (Fig. 9).

In some neurons, including Purkinje cells, a small **persistent** and a **resurgent** Na^+ current (Nav1.6) also help set the subthreshold range between −60–−40 mV. They support high-frequency pacemaking.

36 Intrinsic excitability vs synaptic drive

From your notes (p. 30)



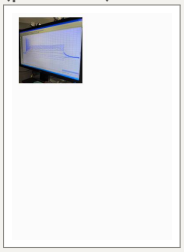
NaV → voltage-gated Na^+/K^+ → intrinsic excitability → action potential. GluR — glutamate receptor; GABA — chloride → puts the cells to action.

This is a very useful way to organize any phenotype:

	Intrinsic excitability	Synaptic drive
Channels	Na_V, K_V, Ca_V, HCN, K_{Ca} (BK/SK)	AMPA, NMDA, $GABA_A$ (ligand-gated)
Question	"Given an input, how does the cell respond?"	"How much input does the cell get?"
Mode	current clamp, K^+ internal	voltage clamp, often Cs^+ internal
Readouts	RMP, R_{in}, rheobase, F–I curve, AP threshold/amplitude/half-width, AHP, sag	sEPSC/mEPSC frequency & amplitude, IPSCs, E/I ratio, AMPA/NMDA ratio

37 Current-clamp step protocol

From your notes (p. 30–31)



Current clamp: at each step in the protocol, current in different amounts. [Screenshot: a family of voltage traces; the largest steps fire a few spikes at onset, then plateau.]

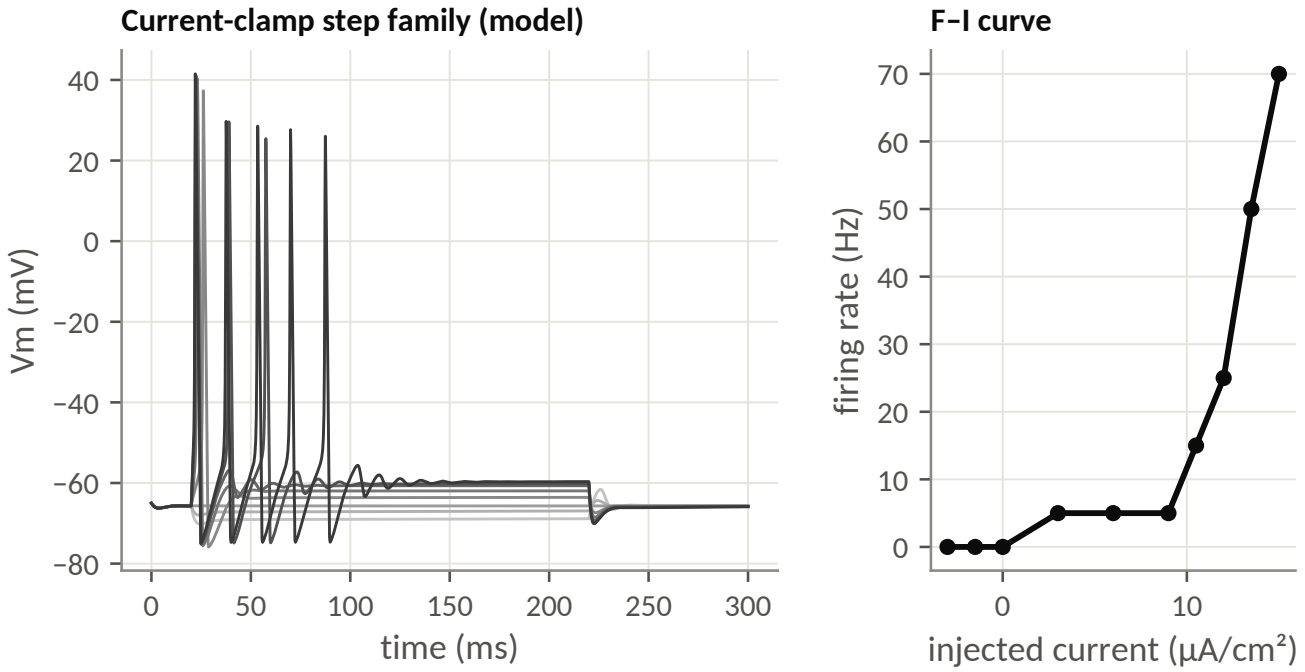


Figure 31: A current-step family and its F–I curve (Hodgkin–Huxley model with a slow adaptation current). Hyperpolarizing steps reveal passive properties. Depolarizing steps reveal the firing pattern.

Reading your screenshot (p. 31). The larger steps fire a burst of spikes at the *start* and then settle on a depolarized plateau. That is **strong spike-frequency adaptation** or **depolarization block**, very common in immature or iPSC-derived neurons, whose Na_V density is still low and which cannot sustain repetitive firing. Tracking this pattern across the astrocyte co-culture groups is a meaningful maturity and excitability readout in itself. More mature neurons fire repetitively for the whole step.

What to extract from a step family:

- **Input resistance** from small hyperpolarizing steps: $R_{\text{in}} = \Delta V / \Delta I$. **Membrane time constant** $\tau_m = R_m C_m$ from the exponential charging at step onset.
- **Sag** during hyperpolarizing steps, which indicates HCN (I_h) channels.
- **Rheobase**: the smallest current step that evokes a spike.
- **AP properties** of the first spike: threshold (where dV/dt exceeds about 10–20 V/s), amplitude, half-width, AHP depth.
- **F–I curve** (spikes per second vs injected current) and **adaptation index** (last inter-spike interval / first).

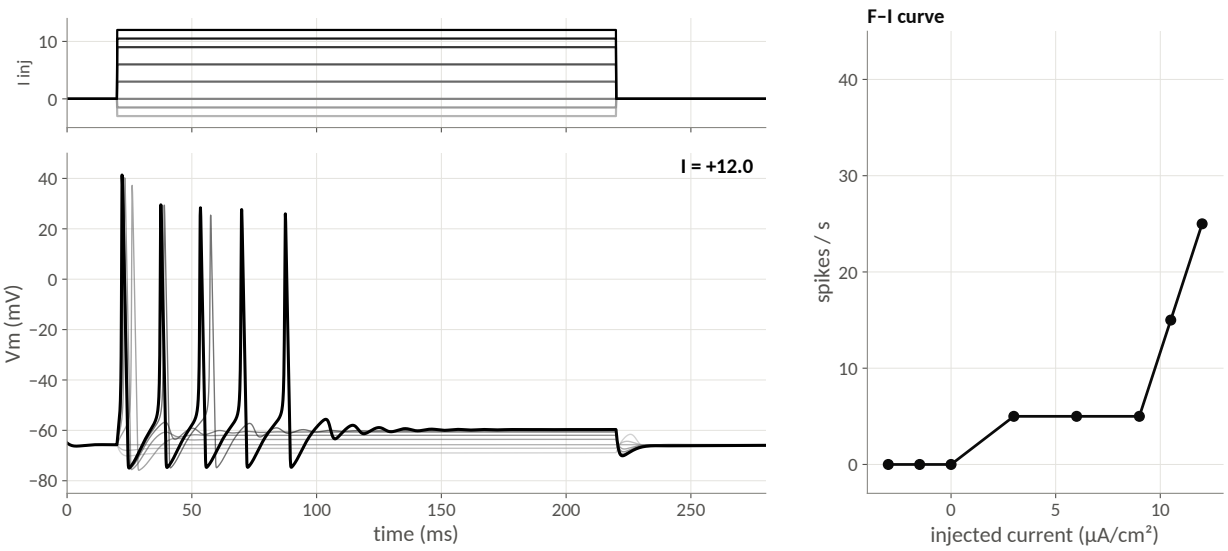


Figure 32: Animation — Current-clamp steps and the F–I curve. Each step is added in turn and its spike count is plotted as the F–I curve. *Press ▶ below the frame (Adobe Acrobat Reader / Okular); other viewers show the final frame. Same animation as video `06_current_clamp_steps.mp4`.*

Appendices

A Quick-reference numbers

Units at the rig

- mV / MΩ = nA; mV / GΩ = pA
- pA × GΩ = mV; MΩ × pF = µs
- $Q = C\,V$: 30 pF × 5 mV = 150 fC
- 1 µF/cm^2 → 10 pF ≈ 1000 µm^2 membrane
- 20 kHz sampling = one point per 50 µs

Typical recording values

- Pipette: 3–7 MΩ
- Seal: $>$ 1 GΩ
- Access R_a: $<$ 20–25 MΩ, stable within 20%
- Cultured neuron C_m: 10–50 pF; R_{in}: 0.2–1 GΩ
- Holding current at −70 mV: small ($>$ −100 pA)
- LJP, K-gluconate: ~15 mV

Reversal potentials (37 °C, typical)

- $E_K \approx -87$ mV; $E_{Na} \approx 60$ mV
- $E_{Cl} \approx -70$ mV (mature); −50–−40 mV (immature)
- E_{glu} (AMPA/NMDA) ≈ 0 mV
- $E_{Ca} \approx 130$ mV

Neuron numbers

- Resting potential: −75–−60 mV
- AP threshold: −55–−40 mV; peak 20–40 mV
- AP half-width: 0.3–2 ms
- AMPA EPSC decay 2–10 ms; NMDA 50–300 ms

B Pharmacology cheat-sheet

Drug	Target	Typical conc.	Use
TTX	Na_V (extracellular)	0.5–1 µM	block spikes → mEPSCs/mIPSCs
QX-314	Na_V (from inside)	5 mM internal	stop escaped spikes in VC
Picrotoxin	$GABA_A$ pore	50–100 µM	isolate EPSCs
Bicuculline	$GABA_A$	10–20 µM	isolate EPSCs
Gabazine (SR-95531)	$GABA_A$	5–10 µM	isolate EPSCs
NBQX / CNQX	AMPA/kainate	10–20 µM	isolate NMDA or IPSCs
D-APV (AP5)	NMDA	50 µM	isolate AMPA; block NMDA-LTP
Kynurenic acid	broad iGluR	1–3 mM	neuroprotection while slicing
TEA	many K_V, BK	1–10 mM	broaden spikes, block K^+
4-AP	K_V1, K_V3 (A-type)	0.1–1 mM	enhance release/excitability
Cs^+ (internal)	K channels	120–135 mM	VC at depolarized potentials
Cs^+ / ZD7288 (external)	HCN (I_h)	2 mM / 10 µM	remove sag

C Transcription notes — things to confirm

I transcribed your handwriting as carefully as possible. These readings are uncertain or looked inconsistent, so please double-check them against what was said in the lab:

1. **p. 2, pipette resistance** read as "6 MΩ". Your screenshot shows 8.7 MΩ; both are within the normal range.
2. **p. 2, glucose** read as 4.6 g; 25 mM is 4.50 g anhydrous (4.95 g monohydrate).
3. **p. 7, metabotropic receptor "SREPB"**: likely "GPCR" (p. 15 confirms G-protein-coupled → PLC).
4. **p. 10, "junction potential: −35"**: unusually large for an LJP; possibly a threshold value (Section 14).
5. **p. 11, "6 [...?]"** next to the groups: possibly the number of coverslips or conditions.
6. **p. 13 & p. 25, calcineurin**: a phosphatase, not a kinase (your description of its effect is right).
7. **p. 17, "I = VR"**: should be I = V/R.
8. **p. 24, "+40 mV to keep Na^+ outside"**: E_{Na} is 60–90 mV; +40 is the AP peak.
9. **p. 26, firing-rate rule** (15–20 Hz vs 10 Hz): confirm which cells and why (it could bias SCA1 comparisons).
10. **p. 28, "I_x current"**: read as the inward (Na^+) current that initiates the spike.

D About the figures, animations and videos

All traces in this guide are **simulations**, not your data. They were made to illustrate the ideas with realistic shapes and magnitudes:

- **Action potentials and voltage-gated currents**: the Hodgkin–Huxley (1952) squid-axon model, shifted to a −65 mV rest, integrated with Euler steps of 5–20 µs. A slow K^+ "adaptation" current was added for the step-family figures, and kinetics were sped up 4× for the sampling example.
- **Seal test / break-in**: an RC circuit model (R_a 15 MΩ, R_m 500 MΩ, C_m 30 pF).
- **NMDA Mg^{2+} block**: Jahr & Stevens (1990).
- **Synaptic events**: difference-of-exponentials conductances with random timing.
- **Nernst/GHK**: textbook ion concentrations.

The nine animations embedded here are also provided as separate MP4 videos (1280×720, with captions), plus one combined "full course" video with title cards:

Video	Topic	Section
`01_resting_potential.mp4`	diffusion vs electrical force, E_K	28
`02_action_potential.mp4`	HH gating, channel cartoons	9
`03_voltage_vs_current_clamp.mp4`	same input, two recording modes	4
`04_seal_test_breakin.mp4`	bath → gigaseal → whole-cell	3
`05_voltage_clamp_iv.mp4`	step protocol → I–V	23
`06_current_clamp_steps.mp4`	step protocol → F–I	37
`07_nmda_mg_block_plasticity.mp4`	Mg^{2+} block, CaMKII, LTP	19
`08_isolating_epsc_ipsc.mp4`	Cs^+ internal, holding potentials	34
`09_sampling_rate.mp4`	digitization	14
`00_ephys_fundamentals_full_course.mp4`	all nine with title cards (~4.5 min)	—

Playing the embedded animations: open this PDF in **Adobe Acrobat Reader** (desktop), or Okular or PDF-XChange. Browser PDF viewers and macOS Preview show only the final frame; use the MP4 files there.

E References and further reading

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- Textbooks: *The Axon Guide* (Molecular Devices) for amplifier and digitizer practice; Hille, *Ion Channels of Excitable Membranes*; Johnston & Wu, *Foundations of Cellular Neurophysiology*.